```
import pandas as pd
import numpy as np
import datetime
from time import strftime
import matplotlib.pyplot as plt
%matplotlib inline

import seaborn as sns

data = pd.read_csv('Data EDA.csv')

data

           PatientId  AppointmentID Gender          ScheduledDay  \
0       2.987250e+13        5642903      F  2016-04-29T18:38:08Z
1       5.589978e+14        5642503      M  2016-04-29T16:08:27Z
2       4.262962e+12        5642549      F  2016-04-29T16:19:04Z
3       8.679512e+11        5642828      F  2016-04-29T17:29:31Z
4       8.841186e+12        5642494      F  2016-04-29T16:07:23Z
...              ...            ...    ...                   ...
110522  2.572134e+12        5651768      F  2016-05-03T09:15:35Z
110523  3.596266e+12        5650093      F  2016-05-03T07:27:33Z
110524  1.557663e+13        5630692      F  2016-04-27T16:03:52Z
110525  9.213493e+13        5630323      F  2016-04-27T15:09:23Z
110526  3.775115e+14        5629448      F  2016-04-27T13:30:56Z

              AppointmentDay  Age      Neighbourhood  Scholarship  \
0       2016-04-29T00:00:00Z   62    JARDIM DA PENHA            0
1       2016-04-29T00:00:00Z   56    JARDIM DA PENHA            0
2       2016-04-29T00:00:00Z   62      MATA DA PRAIA            0
3       2016-04-29T00:00:00Z    8  PONTAL DE CAMBURI            0
4       2016-04-29T00:00:00Z   56    JARDIM DA PENHA            0
...                      ...  ...                ...          ...
110522  2016-06-07T00:00:00Z   56        MARIA ORTIZ            0
110523  2016-06-07T00:00:00Z   51        MARIA ORTIZ            0
110524  2016-06-07T00:00:00Z   21        MARIA ORTIZ            0
110525  2016-06-07T00:00:00Z   38        MARIA ORTIZ            0
110526  2016-06-07T00:00:00Z   54        MARIA ORTIZ            0

        Hipertension  Diabetes  Alcoholism  Handcap  SMS_received No-
show
0                  1         0           0        0             0
No
1                  0         0           0        0             0
No
2                  0         0           0        0             0
No
3                  0         0           0        0             0
No
4                  1         1           0        0             0
```

```
No
...                 ...       ...         ...      ...           ...
...
110522                0         0           0        0             1
No
110523                0         0           0        0             1
No
110524                0         0           0        0             1
No
110525                0         0           0        0             1
No
110526                0         0           0        0             1
No

[110527 rows x 14 columns]
```

```
data.shape
```

```
(110527, 14)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 110527 entries, 0 to 110526
Data columns (total 14 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   PatientId       110527 non-null  float64
 1   AppointmentID   110527 non-null  int64
 2   Gender          110527 non-null  object
 3   ScheduledDay    110527 non-null  object
 4   AppointmentDay  110527 non-null  object
 5   Age             110527 non-null  int64
 6   Neighbourhood   110527 non-null  object
 7   Scholarship     110527 non-null  int64
 8   Hipertension    110527 non-null  int64
 9   Diabetes        110527 non-null  int64
 10  Alcoholism      110527 non-null  int64
 11  Handcap         110527 non-null  int64
 12  SMS_received    110527 non-null  int64
 13  No-show         110527 non-null  object
dtypes: float64(1), int64(8), object(5)
memory usage: 11.8+ MB
```

```
data.describe()
```

```
          PatientId  AppointmentID            Age    Scholarship  \
count  1.105270e+05   1.105270e+05  110527.000000  110527.000000
mean   1.474963e+14   5.675305e+06      37.088874       0.098266
std    2.560949e+14   7.129575e+04      23.110205       0.297675
min    3.921784e+04   5.030230e+06      -1.000000       0.000000
```

```
25%    4.172614e+12   5.640286e+06     18.000000       0.000000
50%    3.173184e+13   5.680573e+06     37.000000       0.000000
75%    9.439172e+13   5.725524e+06     55.000000       0.000000
max    9.999816e+14   5.790484e+06    115.000000       1.000000

        Hipertension       Diabetes    Alcoholism        Handcap  \
count  110527.000000  110527.000000  110527.000000  110527.000000
mean        0.197246       0.071865       0.030400       0.022248
std         0.397921       0.258265       0.171686       0.161543
min         0.000000       0.000000       0.000000       0.000000
25%         0.000000       0.000000       0.000000       0.000000
50%         0.000000       0.000000       0.000000       0.000000
75%         0.000000       0.000000       0.000000       0.000000
max         1.000000       1.000000       1.000000       4.000000

        SMS_received
count  110527.000000
mean        0.321026
std         0.466873
min         0.000000
25%         0.000000
50%         0.000000
75%         1.000000
max         1.000000
```

# Modifying the dataset

```
data['ScheduledDay'] = 
pd.to_datetime(data['ScheduledDay']).dt.date.astype('datetime64[ns]')
data['AppointmentDay'] = 
pd.to_datetime(data['AppointmentDay']).dt.date.astype('datetime64[ns]'
)

data.head(5)
```

```
      PatientId  AppointmentID Gender ScheduledDay AppointmentDay  Age
\
0  2.987250e+13        5642903      F   2016-04-29     2016-04-29   62

1  5.589978e+14        5642503      M   2016-04-29     2016-04-29   56

2  4.262962e+12        5642549      F   2016-04-29     2016-04-29   62

3  8.679512e+11        5642828      F   2016-04-29     2016-04-29    8

4  8.841186e+12        5642494      F   2016-04-29     2016-04-29   56


      Neighbourhood  Scholarship  Hipertension  Diabetes  Alcoholism
\
```

```
0    JARDIM DA PENHA            0             1         0           0

1    JARDIM DA PENHA            0             0         0           0

2      MATA DA PRAIA            0             0         0           0

3  PONTAL DE CAMBURI            0             0         0           0

4    JARDIM DA PENHA            0             1         1           0


   Handcap  SMS_received No-show
0        0             0      No
1        0             0      No
2        0             0      No
3        0             0      No
4        0             0      No
```

## for ScheduledDay & AppointmentDay lets store weekdays in a variable

```
data['s_wday'] = data['ScheduledDay'].dt.dayofweek

data['ap_wday'] = data['AppointmentDay'].dt.dayofweek

data['s_wday'].value_counts()

s_wday
1    26168
2    24262
0    23085
4    18915
3    18073
5       24
Name: count, dtype: int64

data['ap_wday'].value_counts()

ap_wday
2    25867
1    25640
0    22715
4    19019
3    17247
5       39
Name: count, dtype: int64

data.columns
```

```
Index(['PatientId', 'AppointmentID', 'Gender', 'ScheduledDay',
       'AppointmentDay', 'Age', 'Neighbourhood', 'Scholarship',
'Hipertension',
       'Diabetes', 'Alcoholism', 'Handcap', 'SMS_received', 'No-show',
       's_wday', 'ap_wday'],
      dtype='object')
```

# Modifying the dataset by giving proper column names

```
data= data.rename(columns={'Hipertension': 'Hypertension', 'Handcap':
'Handicap', 'SMS_received': 'SMSReceived', 'No-show': 'NoShow'})

data.columns

Index(['PatientId', 'AppointmentID', 'Gender', 'ScheduledDay',
       'AppointmentDay', 'Age', 'Neighbourhood', 'Scholarship',
'Hypertension',
       'Diabetes', 'Alcoholism', 'Handicap', 'SMSReceived', 'NoShow',
's_wday',
       'ap_wday'],
      dtype='object')
```

## Removing some columns which are not important

```
data.drop(['AppointmentID', 'Neighbourhood','PatientId'], axis=1,
inplace=True)

data.head(5)

  Gender ScheduledDay AppointmentDay  Age  Scholarship
Hypertension  \
0      F   2016-04-29     2016-04-29   62            0            1

1      M   2016-04-29     2016-04-29   56            0            0

2      F   2016-04-29     2016-04-29   62            0            0

3      F   2016-04-29     2016-04-29    8            0            0

4      F   2016-04-29     2016-04-29   56            0            1


   Diabetes  Alcoholism  Handicap  SMSReceived NoShow  s_wday  ap_wday

0         0           0         0            0     No       4        4

1         0           0         0            0     No       4        4

2         0           0         0            0     No       4        4
```

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 0 | 0 | 0 | 0 | No | 4 | 4 |
| 4 | 1 | 0 | 0 | 0 | No | 4 | 4 |

```
data.describe()
```

| | ScheduledDay | AppointmentDay |
|---|---|---|
| count | 110527 | 110527 |
| mean | 2016-05-08 20:33:18.179630080 | 2016-05-19 00:57:50.008233472 |
| min | 2015-11-10 00:00:00 | 2016-04-29 00:00:00 |
| 25% | 2016-04-29 00:00:00 | 2016-05-09 00:00:00 |
| 50% | 2016-05-10 00:00:00 | 2016-05-18 00:00:00 |
| 75% | 2016-05-20 00:00:00 | 2016-05-31 00:00:00 |
| max | 2016-06-08 00:00:00 | 2016-06-08 00:00:00 |
| std | NaN | NaN |

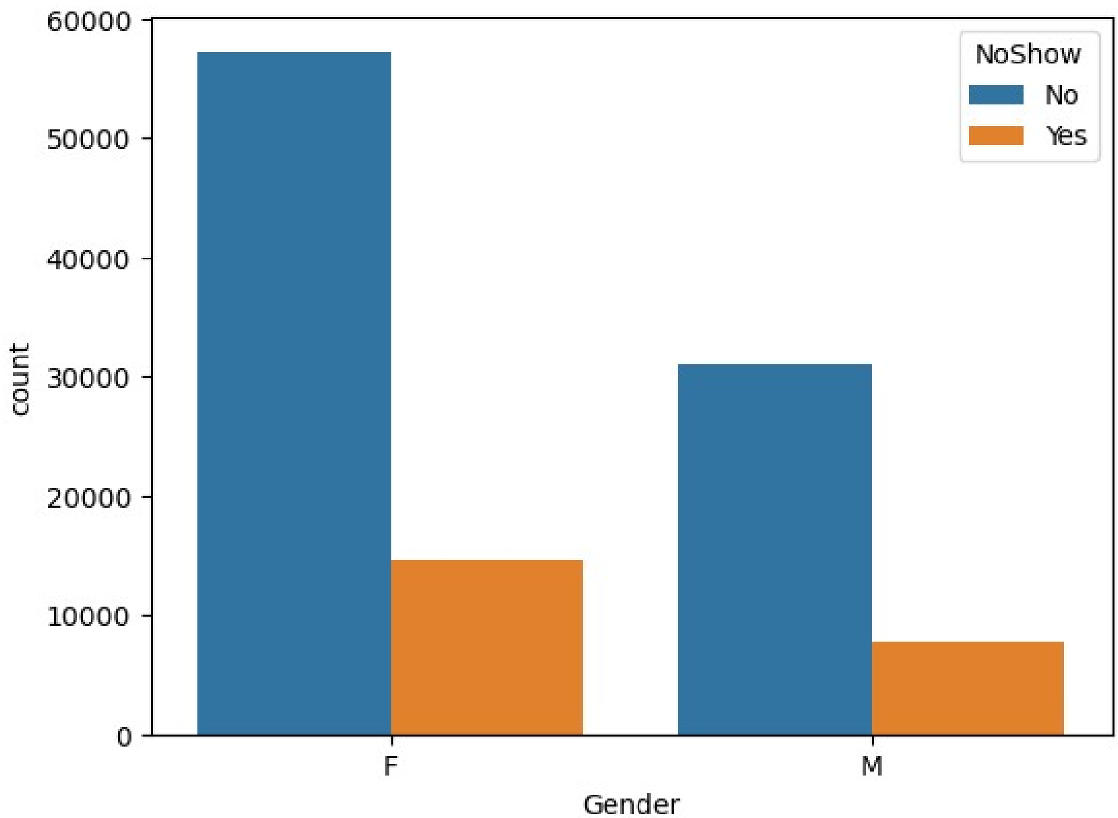
| | Age | Scholarship | Hypertension | Diabetes |
|---|---|---|---|---|
| count | 110527.000000 | 110527.000000 | 110527.000000 | 110527.000000 |
| mean | 37.088874 | 0.098266 | 0.197246 | 0.071865 |
| min | -1.000000 | 0.000000 | 0.000000 | 0.000000 |
| 25% | 18.000000 | 0.000000 | 0.000000 | 0.000000 |
| 50% | 37.000000 | 0.000000 | 0.000000 | 0.000000 |
| 75% | 55.000000 | 0.000000 | 0.000000 | 0.000000 |
| max | 115.000000 | 1.000000 | 1.000000 | 1.000000 |
| std | 23.110205 | 0.297675 | 0.397921 | 0.258265 |

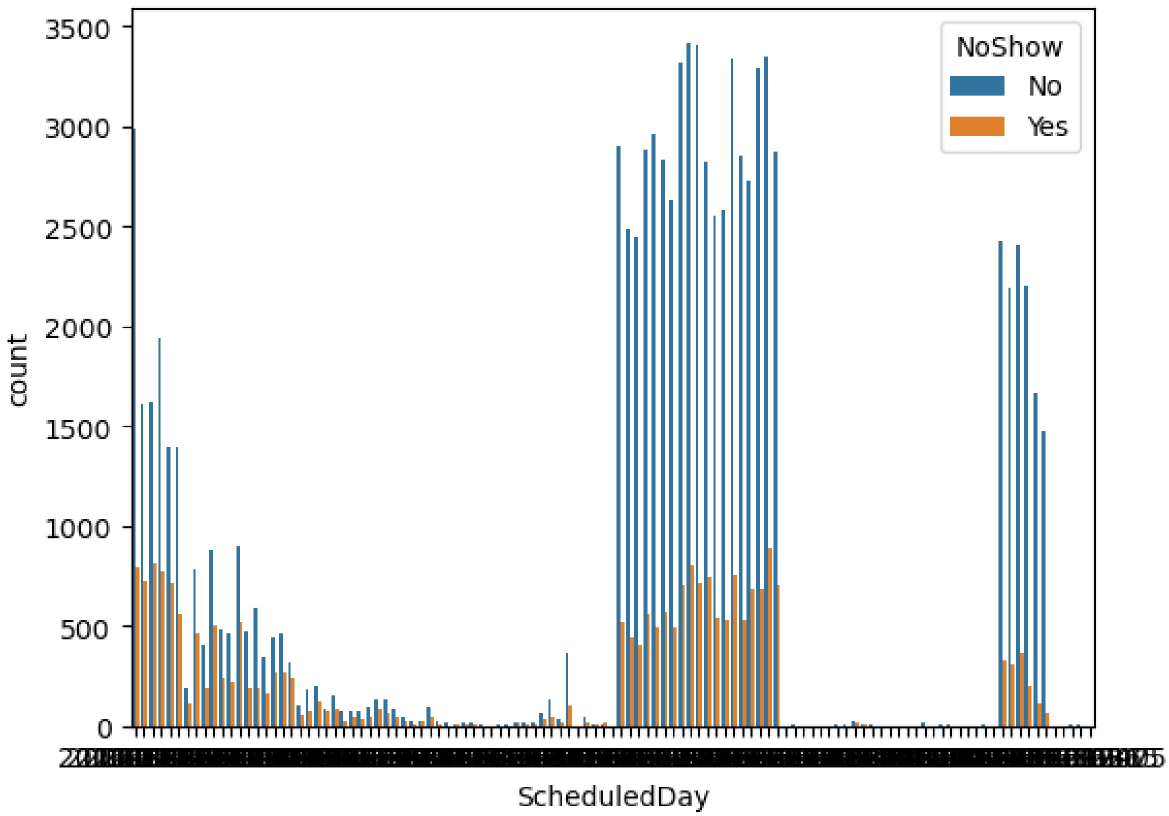
| | Alcoholism | Handicap | SMSReceived | s_wday |
|---|---|---|---|---|
| count | 110527.000000 | 110527.000000 | 110527.000000 | 110527.000000 |
| mean | 0.030400 | 0.022248 | 0.321026 | 1.851955 |
| min | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| 25% | 0.000000 | 0.000000 | 0.000000 | 1.000000 |
| 50% | 0.000000 | 0.000000 | 0.000000 | 2.000000 |
| 75% | 0.000000 | 0.000000 | 1.000000 | 3.000000 |
| max | 1.000000 | 4.000000 | 1.000000 | 5.000000 |
| std | 0.171686 | 0.161543 | 0.466873 | 1.378520 |

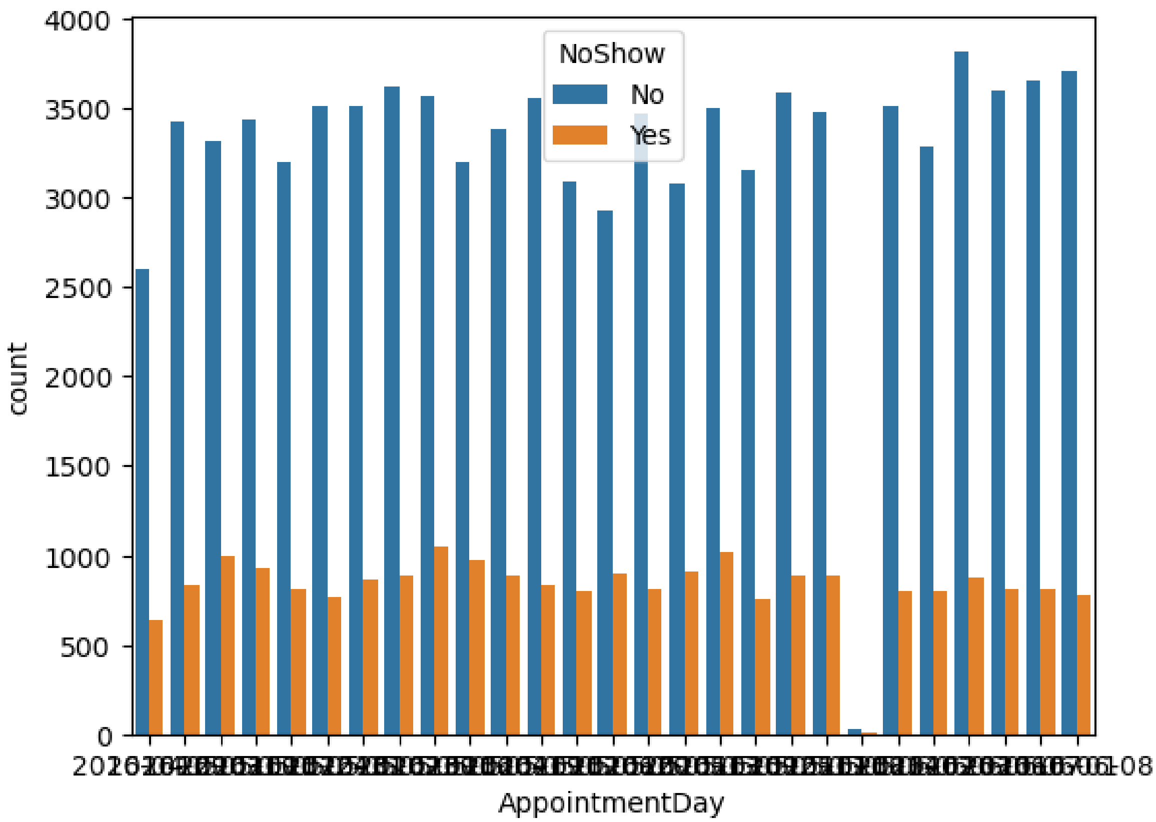
| | ap_wday |
|---|---|
| count | 110527.000000 |
| mean | 1.858243 |
| min | 0.000000 |
| 25% | 1.000000 |
| 50% | 2.000000 |
| 75% | 3.000000 |
| max | 5.000000 |
| std | 1.371672 |

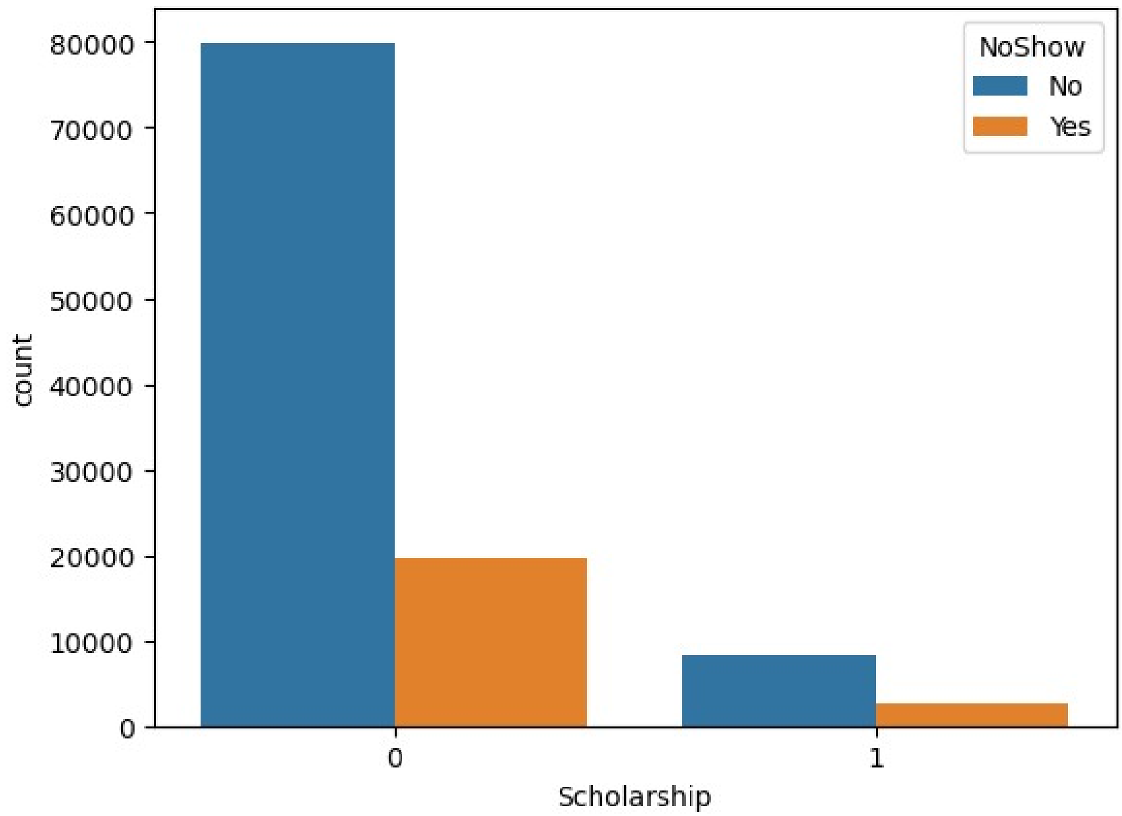
# Check for Distribution of Appointment No-Shows

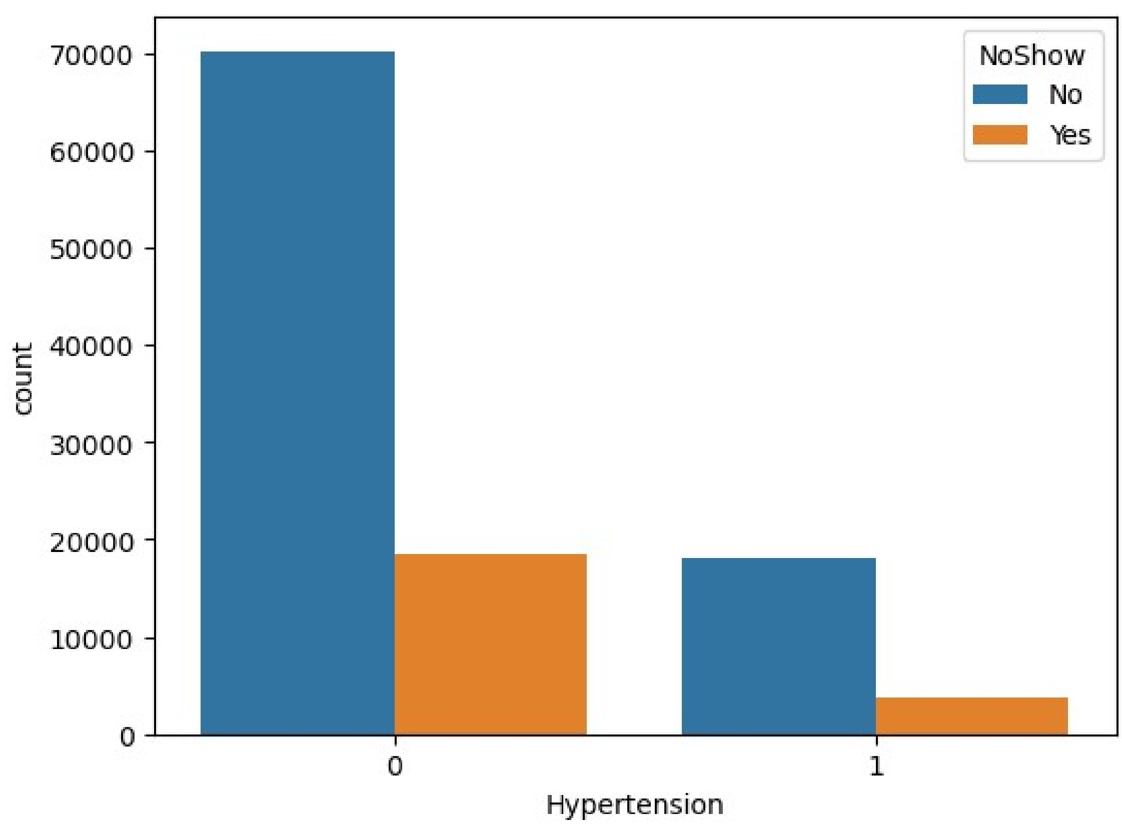
```
data['NoShow'].value_counts().plot(kind='barh', figsize=(8, 6))
plt.xlabel("Count", labelpad=14)
plt.ylabel("Target Variable", labelpad=14)
plt.title("Count of TARGET Variable per category", y=1.02);
plt.show()
```

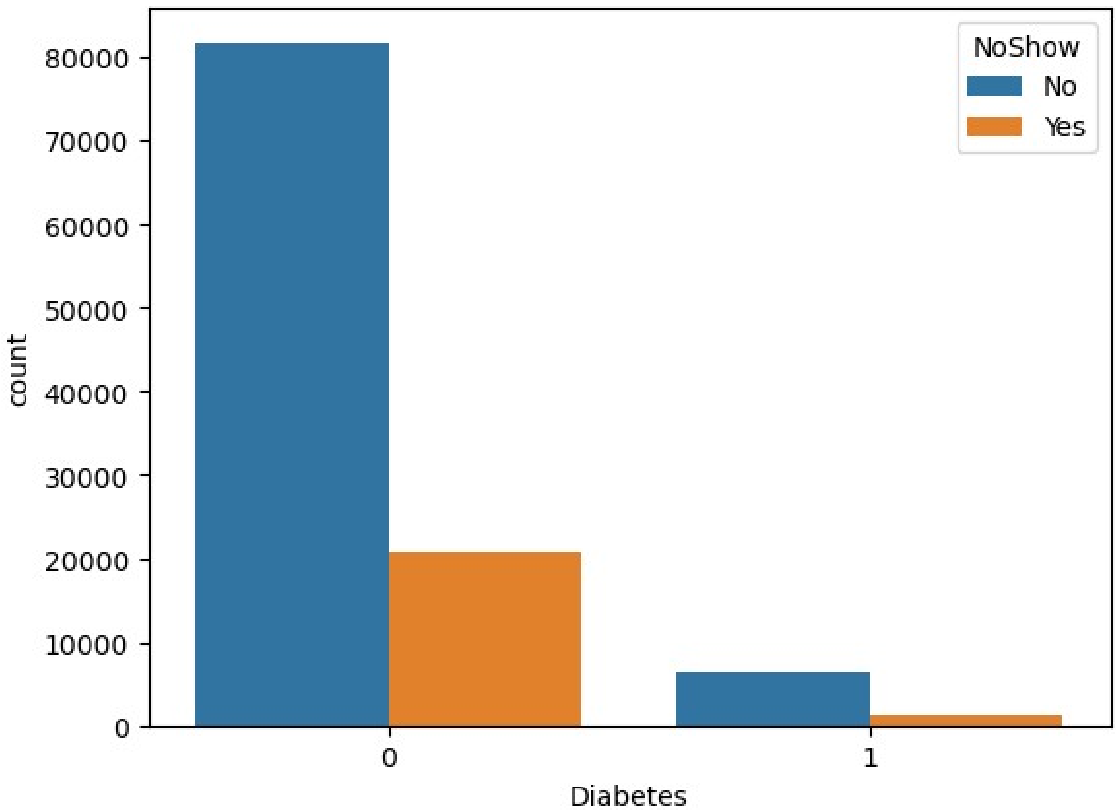


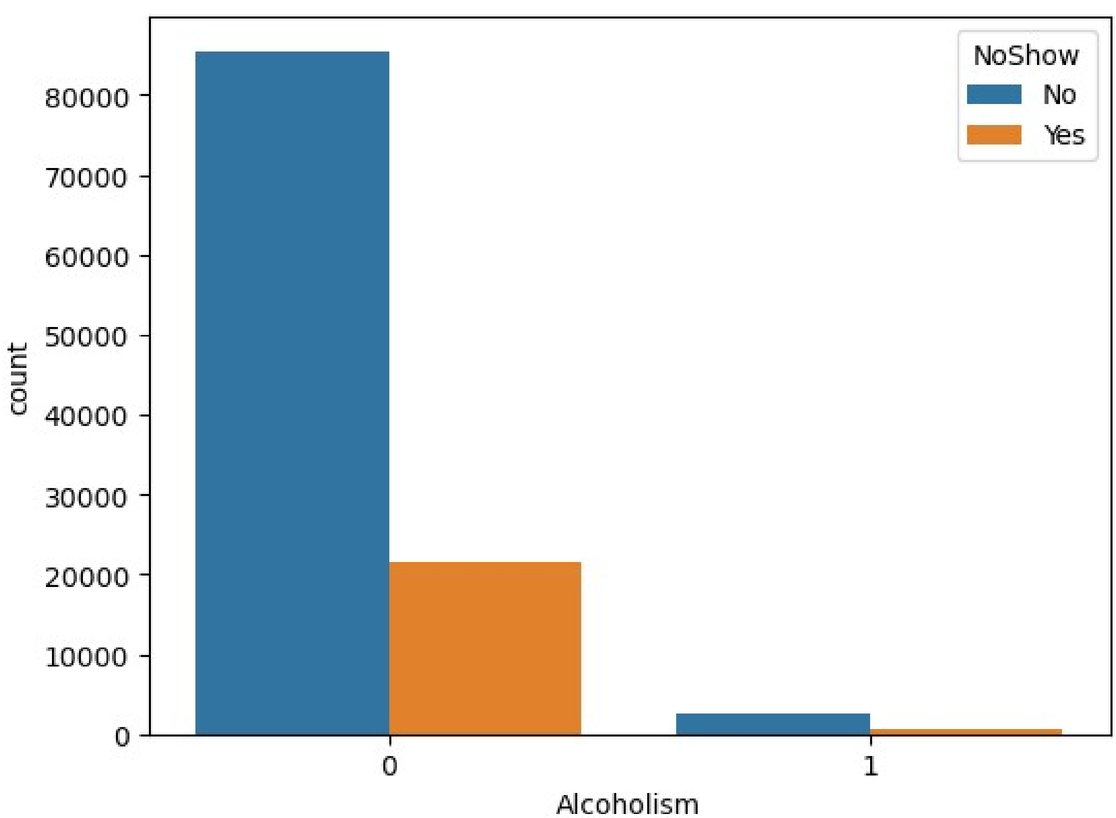

NoShow
No
Yes
count
0
500
1000
1500
2000
2500
3000
3500
ScheduledDay

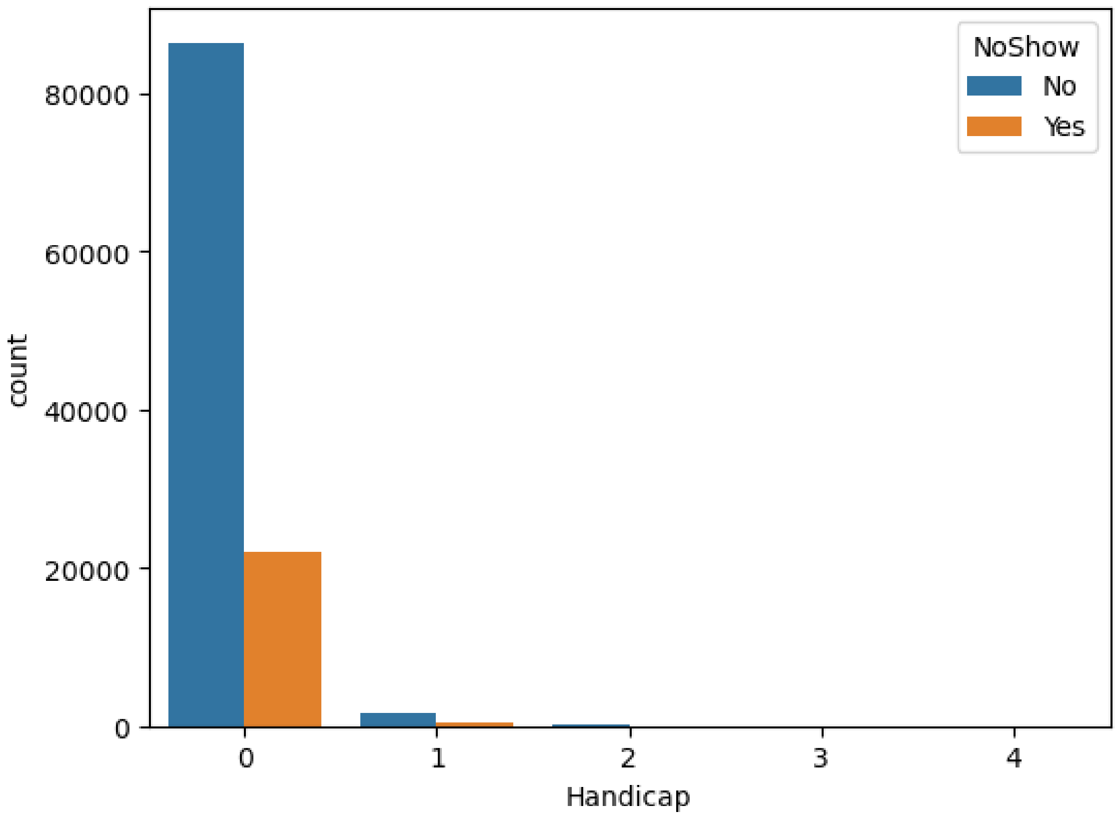

NoShow
No
Yes
count
0
500
1000
1500
2000
2500
3000
3500
4000
AppointmentDay

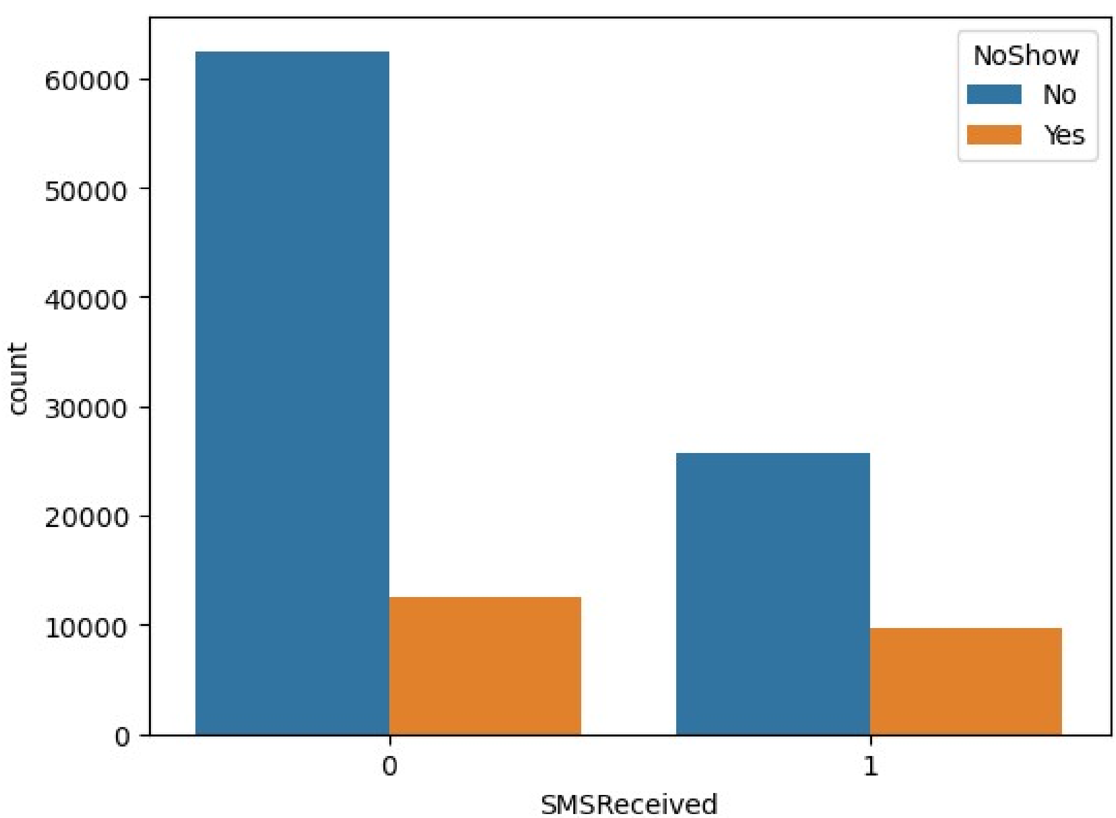

NoShow
No
Yes
count
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Scholarship

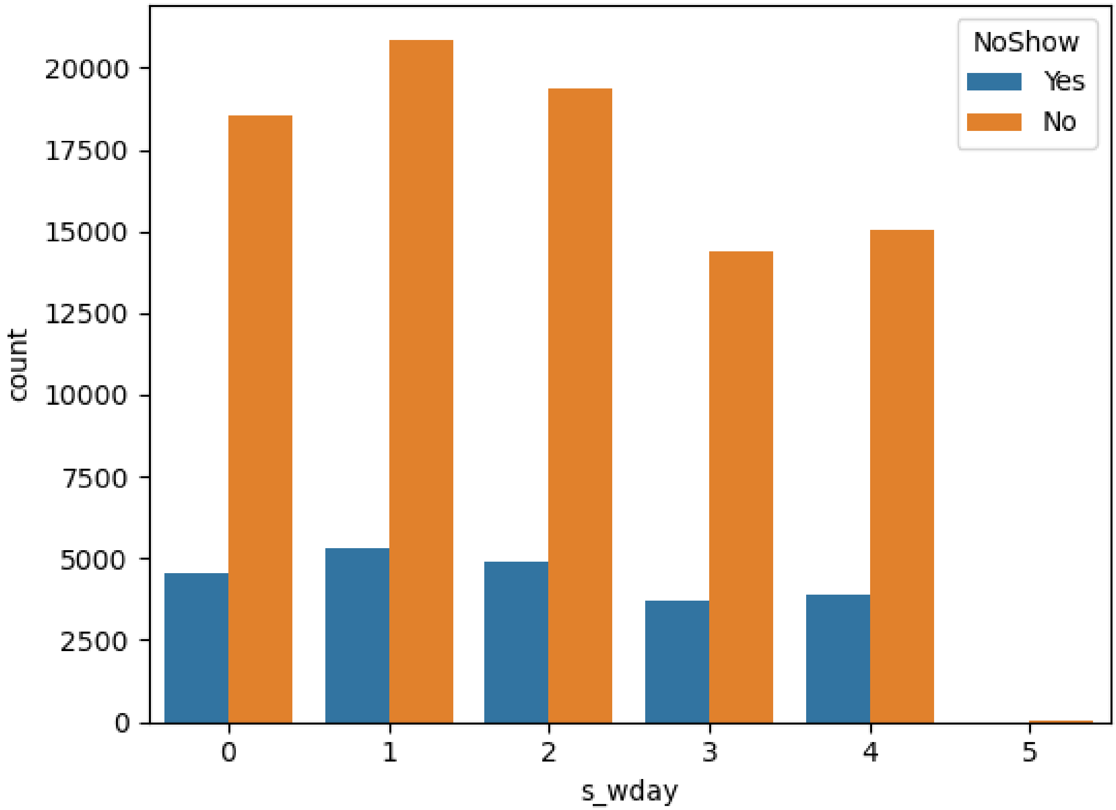

NoShow
No
Yes
count
70000
60000
50000
40000
30000
20000
10000
0
0
1
Hypertension

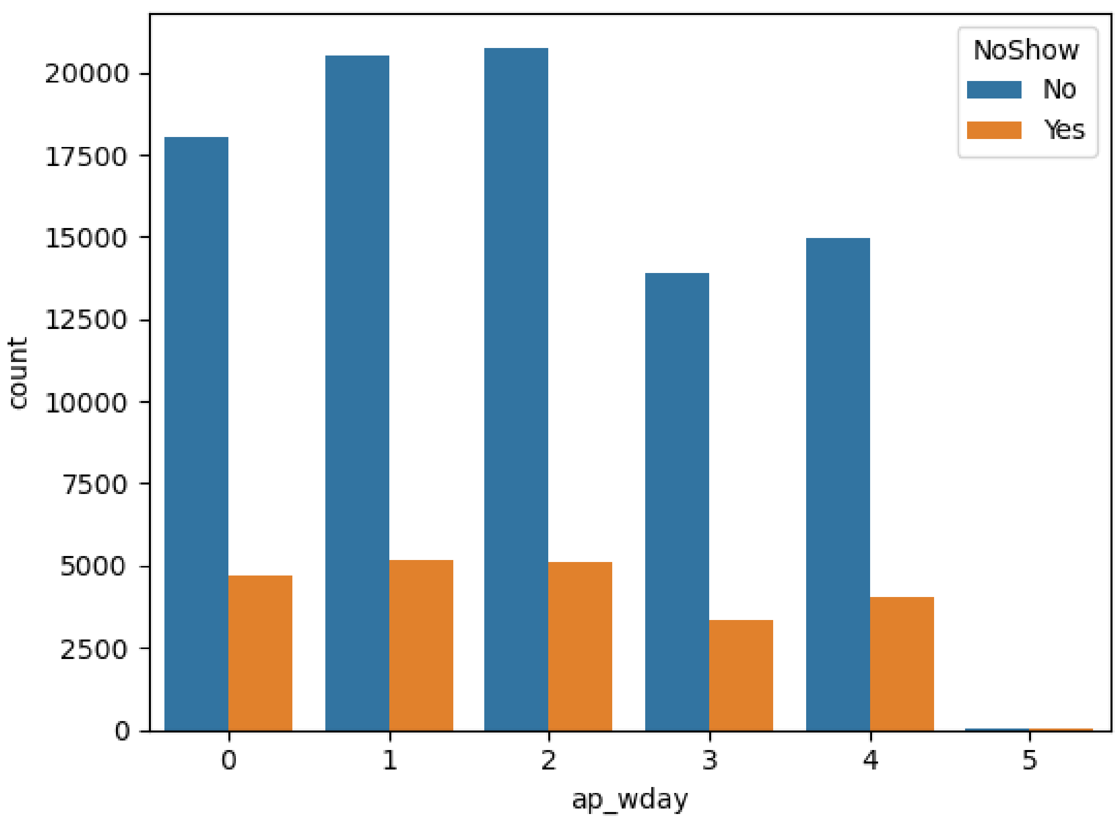

NoShow
No
Yes
count
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Diabetes

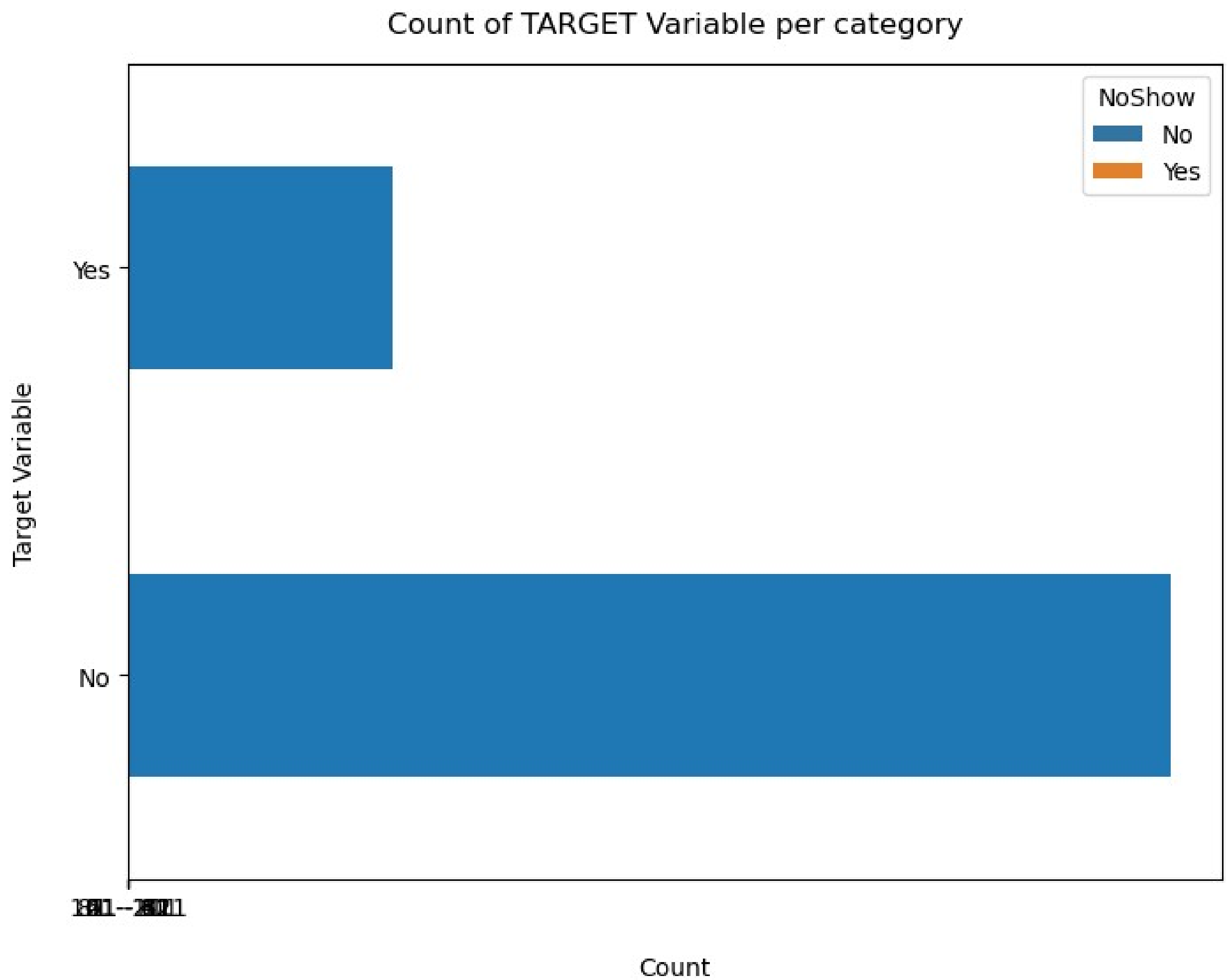

NoShow
No
Yes
count
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Alcoholism


NoShow
No
Yes
count
80000
60000
40000
20000
0
0
1
2
3
4
Handicap


NoShow
No
Yes
count
60000
50000
40000
30000
20000
10000
0
0
1
SMSReceived


NoShow
Yes
No
count
s_wday
0
1
2
3
4
5
0
2500
5000
7500
10000
12500
15000
17500
20000


NoShow
No
Yes
count
ap_wday
0
1
2
3
4
5
0
2500
5000
7500
10000
12500
15000
17500
20000




#The "NoShow" (Yes) category represents a smaller portion of the dataset.

## Exact percentage of no-shows

```
100*data['NoShow'].value_counts()/len(data['NoShow'])

NoShow
No     79.806744
Yes    20.193256
Name: count, dtype: float64
```

#The dataset is imbalanced, with 79.8% of patients attending their appointments ("No") and 20.2% missing them ("Yes").

## Having a look that data contains missing values or not

```
missing = pd.DataFrame((data.isnull().sum()) * 100 /
data.shape[0]).reset_index()
plt.figure(figsize=(16, 5))
ax = sns.pointplot(x='index', y=0, data=missing)
```

```
plt.xticks(rotation=90, fontsize=7)
plt.title("Percentage of Missing values")
plt.ylabel("PERCENTAGE")
plt.show()
```


#No missing values.

# DATA Cleaning

```
new_data=data.copy()

new_data.head(2)

  Gender ScheduledDay AppointmentDay  Age  Scholarship  
Hypertension  \
0      F   2016-04-29     2016-04-29   62            0              1

1      M   2016-04-29     2016-04-29   56            0              0


   Diabetes  Alcoholism  Handicap  SMSReceived NoShow  s_wday  ap_wday

0         0           0         0            0     No       4        4

1         0           0         0            0     No       4        4


new_data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 110527 entries, 0 to 110526
Data columns (total 13 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
```

```
 0   Gender          110527 non-null  object
 1   ScheduledDay    110527 non-null  datetime64[ns]
 2   AppointmentDay  110527 non-null  datetime64[ns]
 3   Age             110527 non-null  int64
 4   Scholarship     110527 non-null  int64
 5   Hypertension    110527 non-null  int64
 6   Diabetes        110527 non-null  int64
 7   Alcoholism      110527 non-null  int64
 8   Handicap        110527 non-null  int64
 9   SMSReceived     110527 non-null  int64
 10  NoShow          110527 non-null  object
 11  s_wday          110527 non-null  int32
 12  ap_wday         110527 non-null  int32
dtypes: datetime64[ns](2), int32(2), int64(7), object(2)
memory usage: 10.1+ MB
```

#Since there are no null records in this dataset , no cleaning is required further

# Grouping the numerical Age column into categorical buckets (bins) of 20-year intervals.

```python
# The max tenure -
print(data['Age'].max())
```

```
115
```

```python
labels = ["{0} - {1}".format(i, i + 20) for i in range(1, 118, 20)]
data['Age_group'] = pd.cut(data.Age, range(1, 130, 20), right=False,
labels=labels)

data.drop(['Age'], axis=1, inplace=True)

data
```

```
       Gender ScheduledDay AppointmentDay  Scholarship
Hypertension  \
0           F   2016-04-29     2016-04-29            0             1

1           M   2016-04-29     2016-04-29            0             0

2           F   2016-04-29     2016-04-29            0             0

3           F   2016-04-29     2016-04-29            0             0

4           F   2016-04-29     2016-04-29            0             1

...       ...          ...            ...          ...           ...

110522      F   2016-05-03     2016-06-07            0             0
```

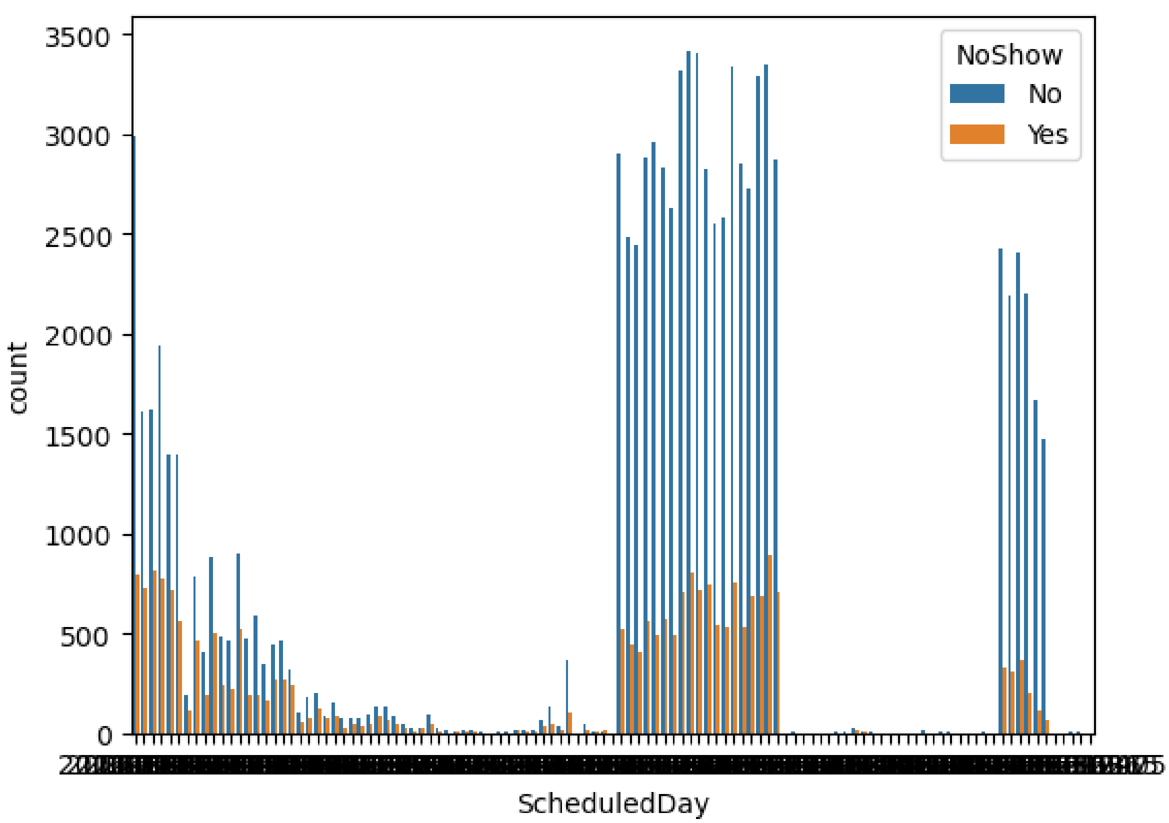
```
110523      F   2016-05-03     2016-06-07           0             0

110524      F   2016-04-27     2016-06-07           0             0

110525      F   2016-04-27     2016-06-07           0             0

110526      F   2016-04-27     2016-06-07           0             0


        Diabetes  Alcoholism  Handicap  SMSReceived NoShow  s_wday
ap_wday  \
0              0           0         0            0     No       4
4
1              0           0         0            0     No       4
4
2              0           0         0            0     No       4
4
3              0           0         0            0     No       4
4
4              1           0         0            0     No       4
4
...          ...         ...       ...          ...    ...     ...
...
110522         0           0         0            1     No       1
1
110523         0           0         0            1     No       1
1
110524         0           0         0            1     No       2
1
110525         0           0         0            1     No       2
1
110526         0           0         0            1     No       2
1

       Age_group
0        61 - 81
1        41 - 61
2        61 - 81
3         1 - 21
4        41 - 61
...          ...
110522   41 - 61
110523   41 - 61
110524   21 - 41
110525   21 - 41
110526   41 - 61

[110527 rows x 13 columns]
```

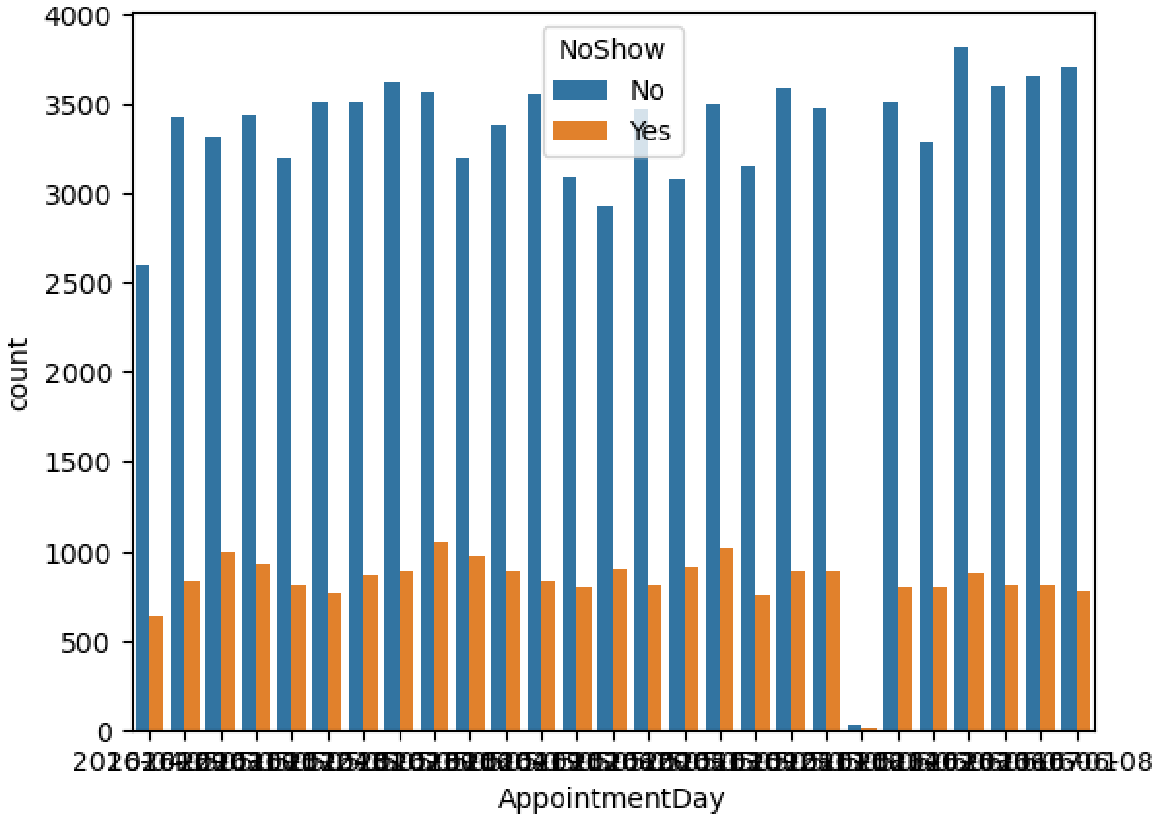
# DATA Exploration

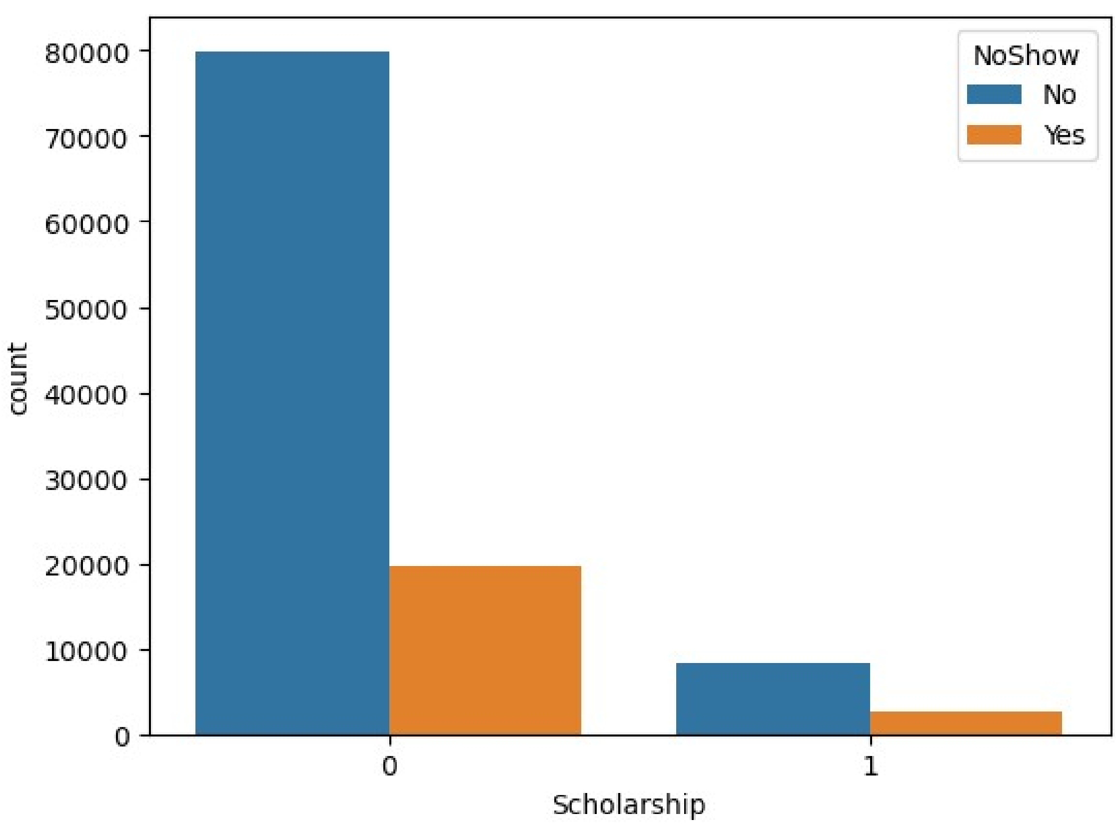
Bivariate analysis by looping through all features to display their frequency counts and visualize their relationship with the NoShow target variable using side-by-side count plots.

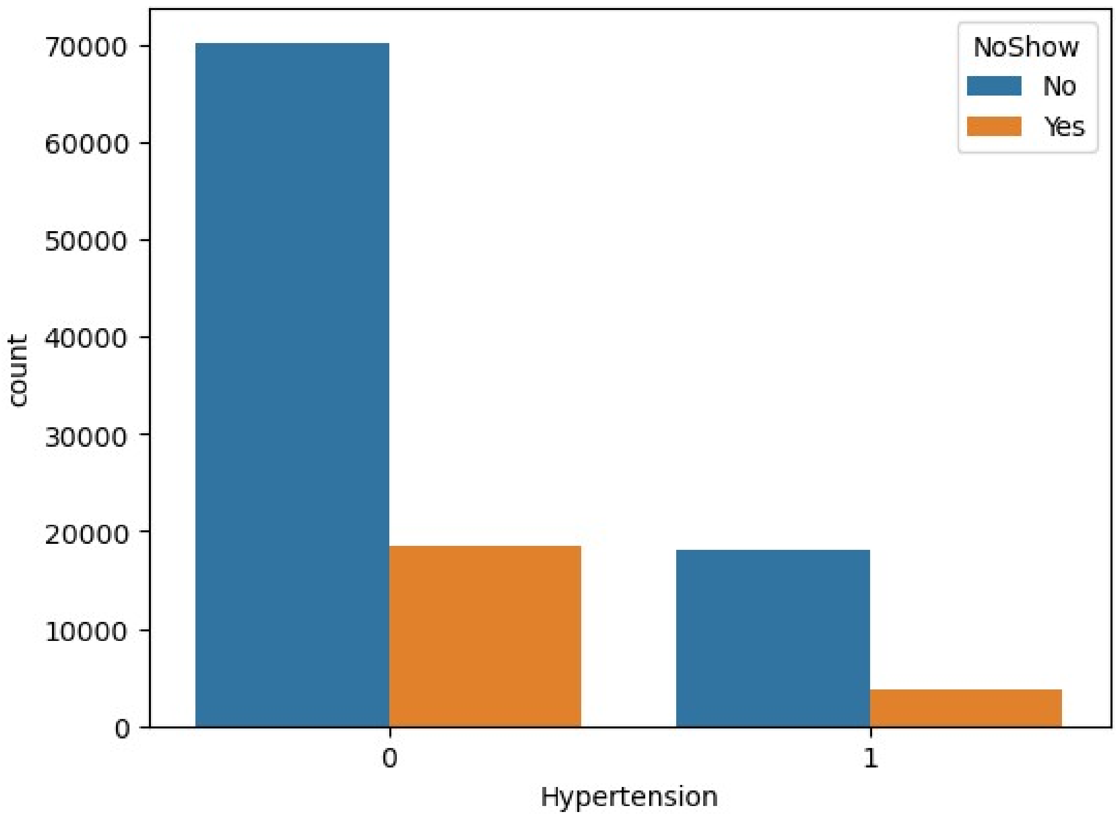
```
for i, predictor in enumerate(data.drop(columns=['NoShow'])):
    print('-'*10,predictor,'-'*10)
    print(data[predictor].value_counts())
    plt.figure(i)
    sns.countplot(data=data, x=predictor, hue='NoShow')
    plt.show()
```

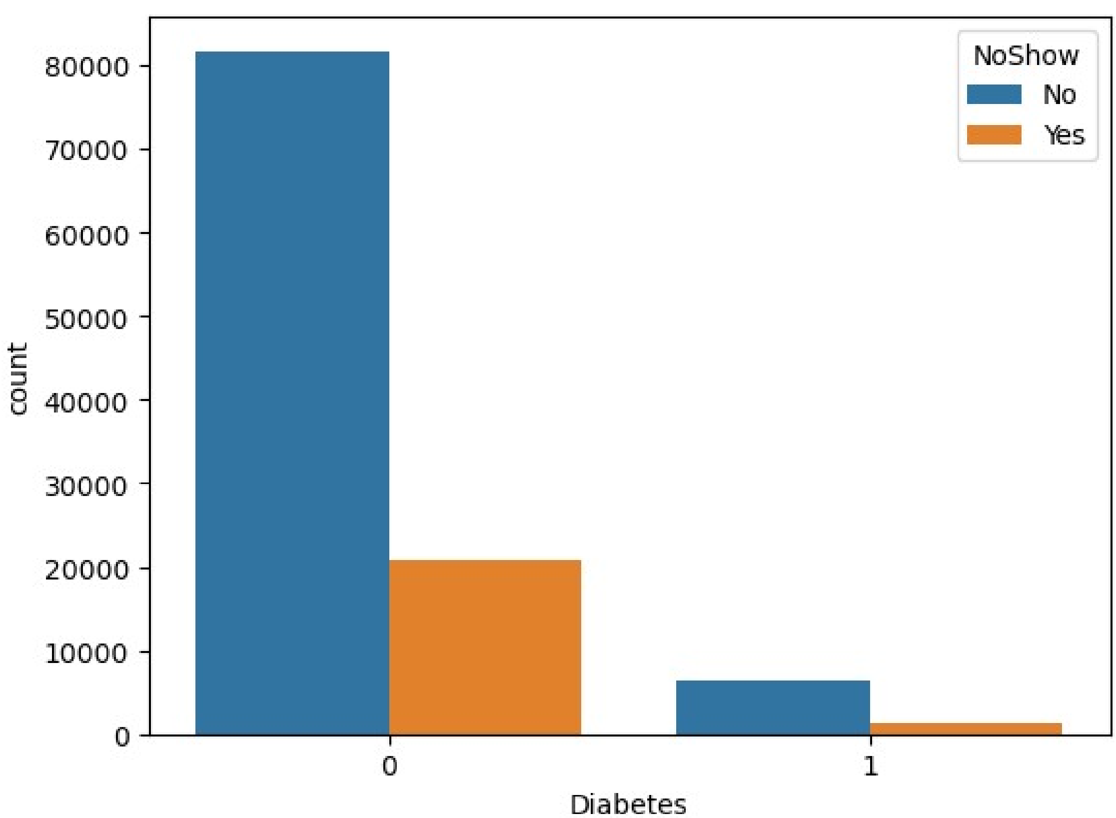
```
---------- Gender ----------
Gender
F    71840
M    38687
Name: count, dtype: int64
```

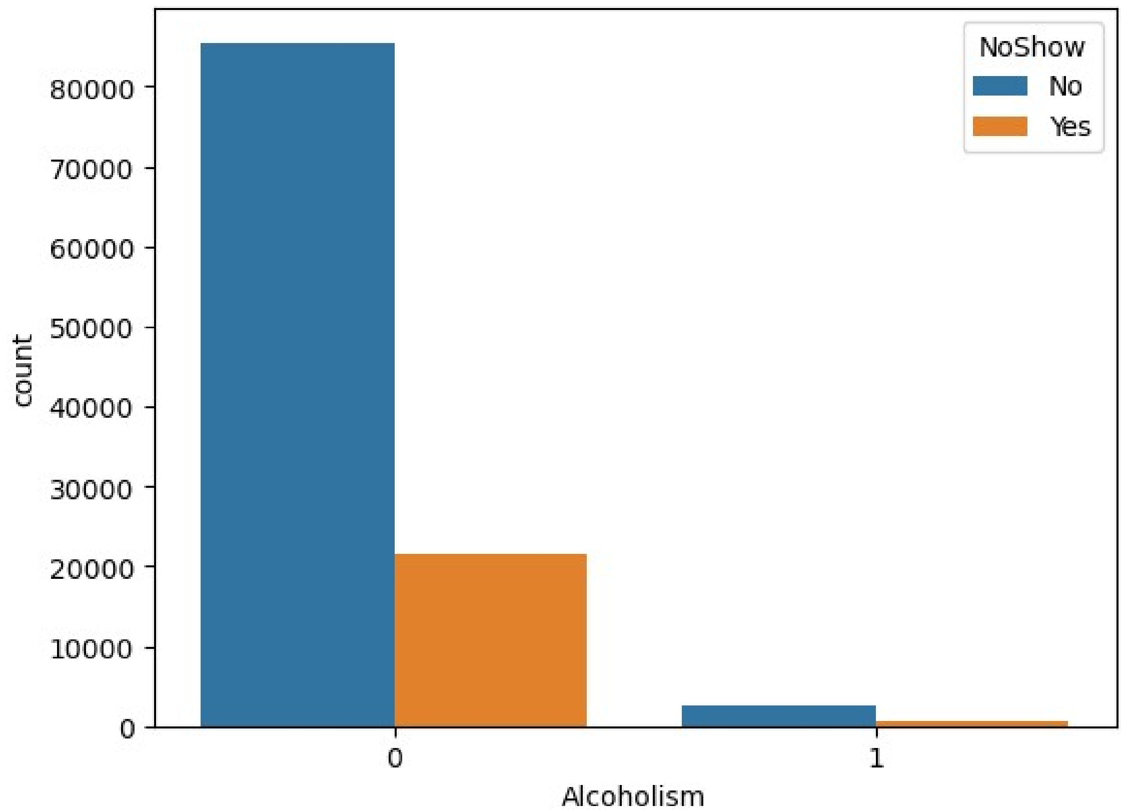


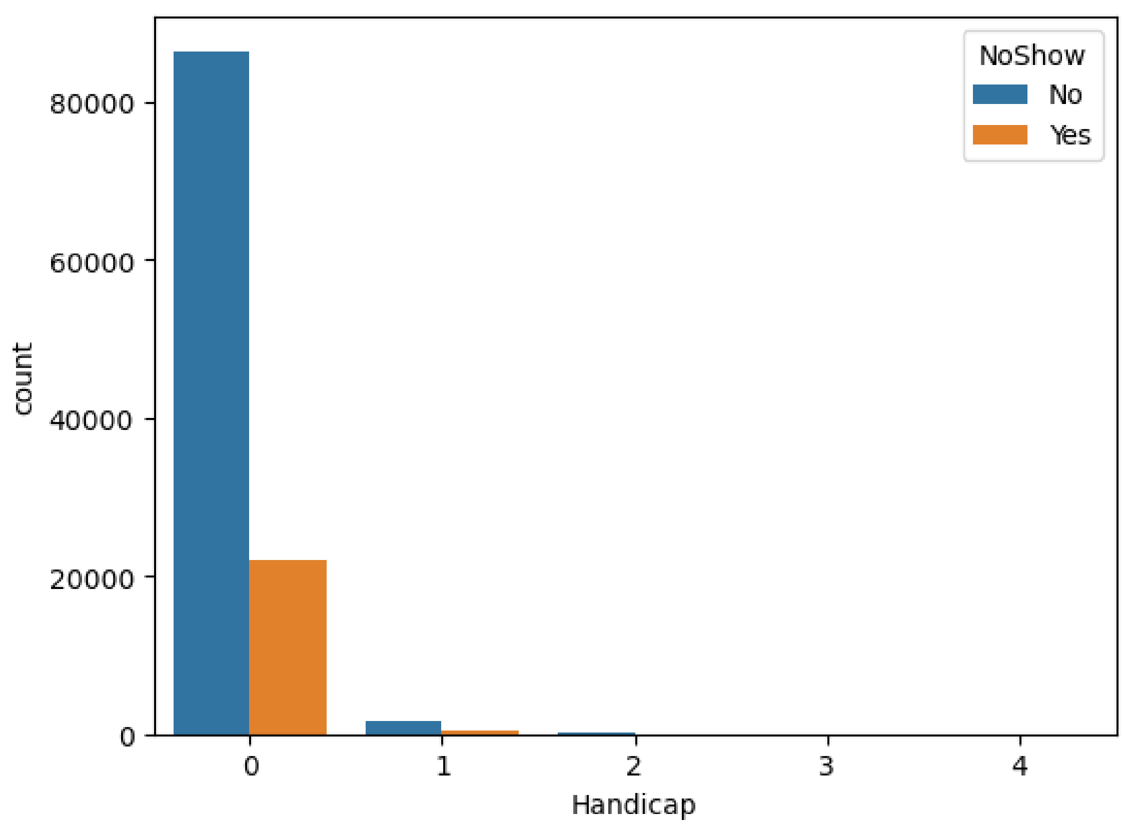

4000
3500
3000
2500
2000
1500
1000
500
0
count
NoShow
No
Yes
AppointmentDay

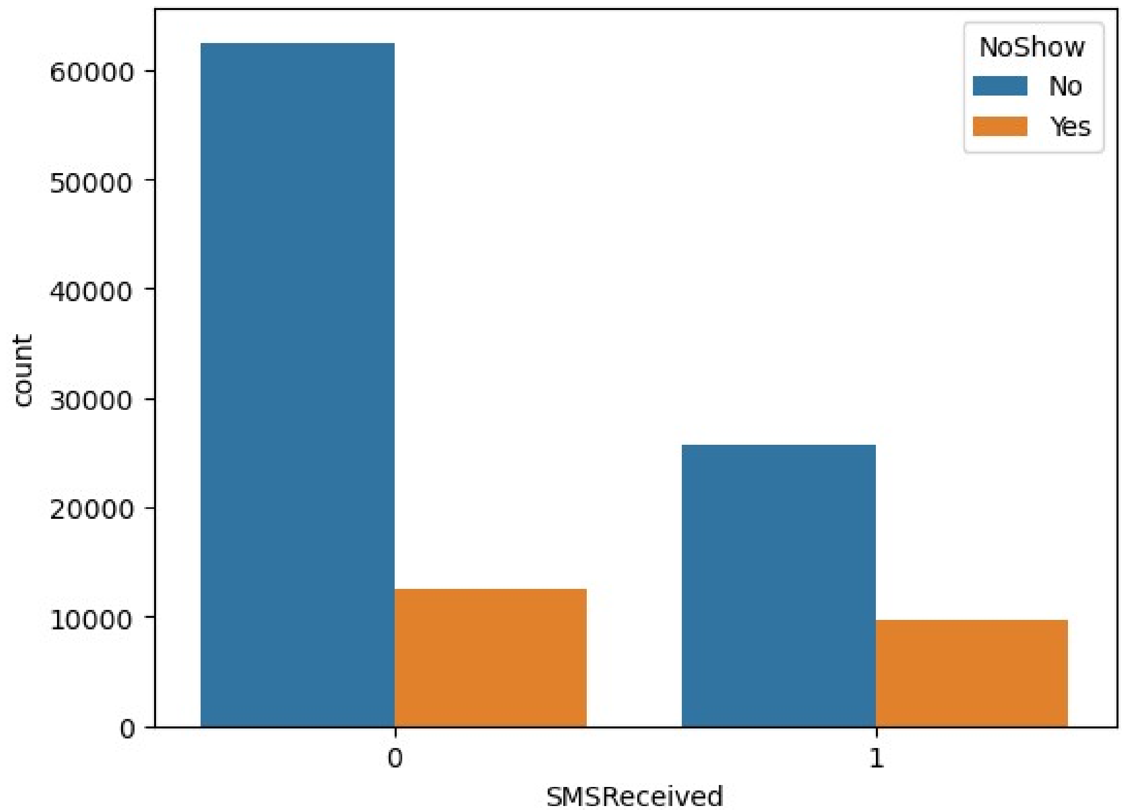

80000
70000
60000
50000
40000
30000
20000
10000
0
count
NoShow
No
Yes
0
1
Scholarship

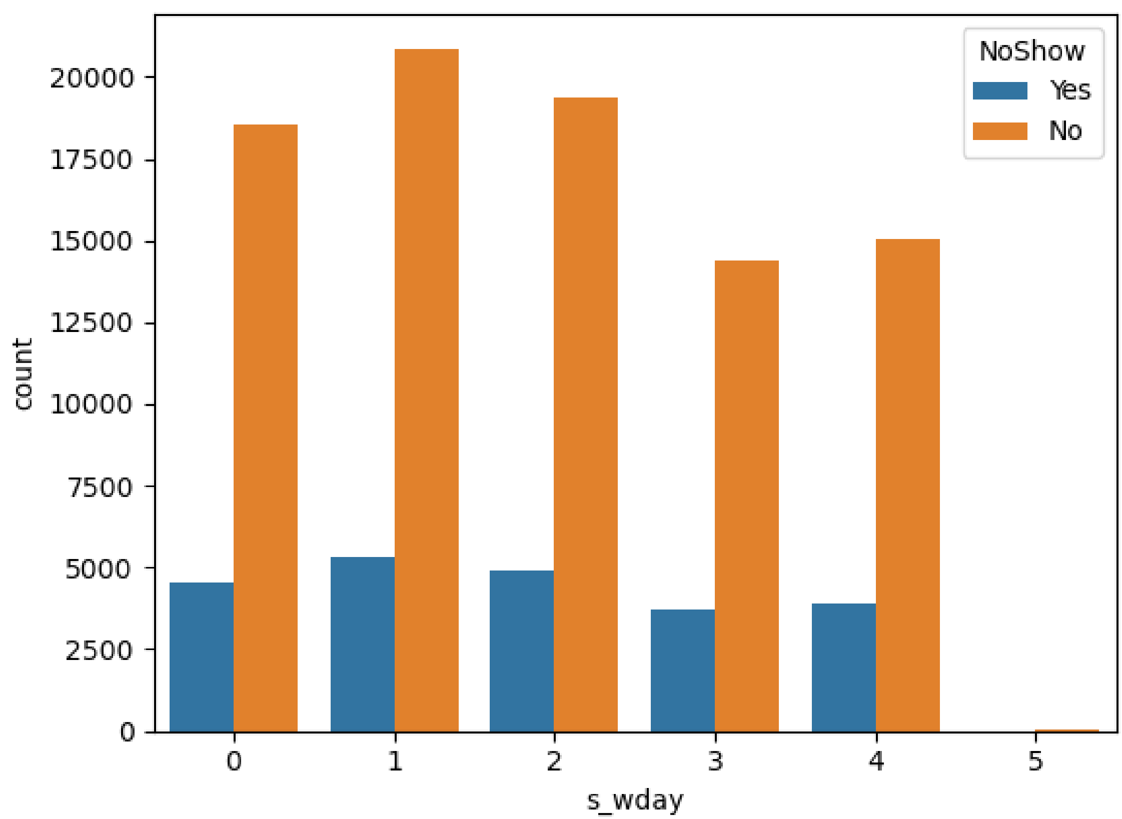

NoShow
No
Yes
count
70000
60000
50000
40000
30000
20000
10000
0
0
1
Hypertension

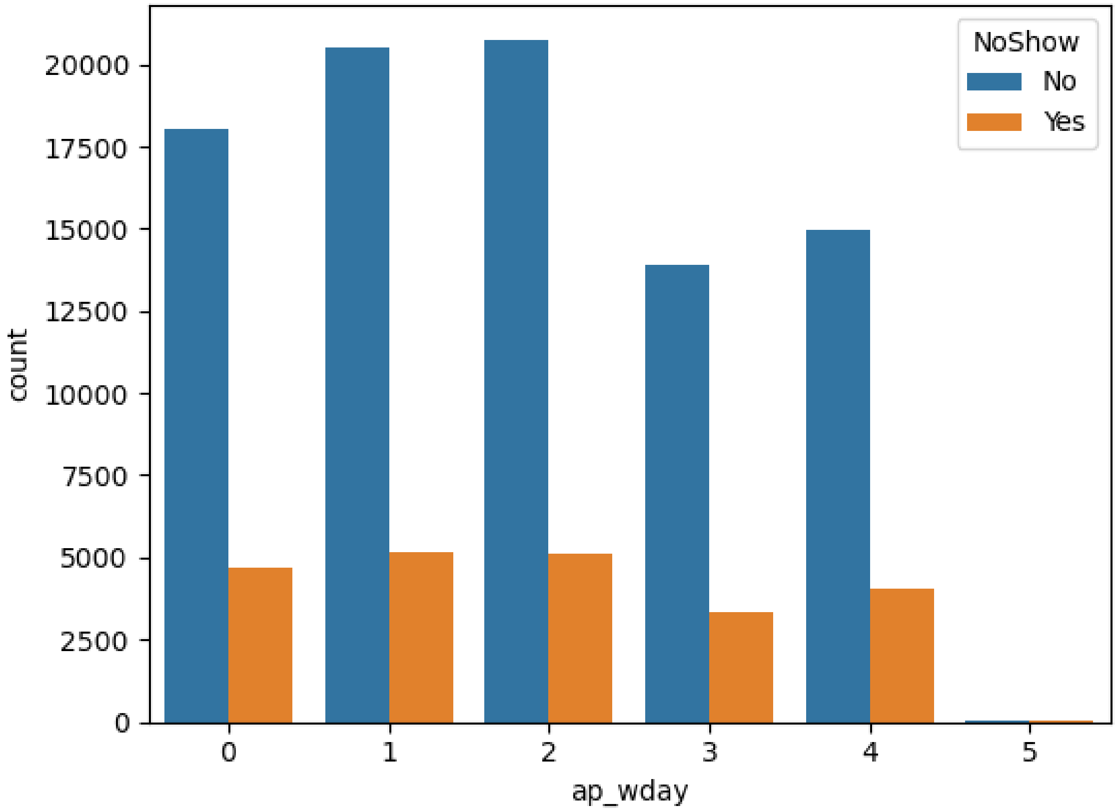

NoShow
No
Yes
count
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Diabetes

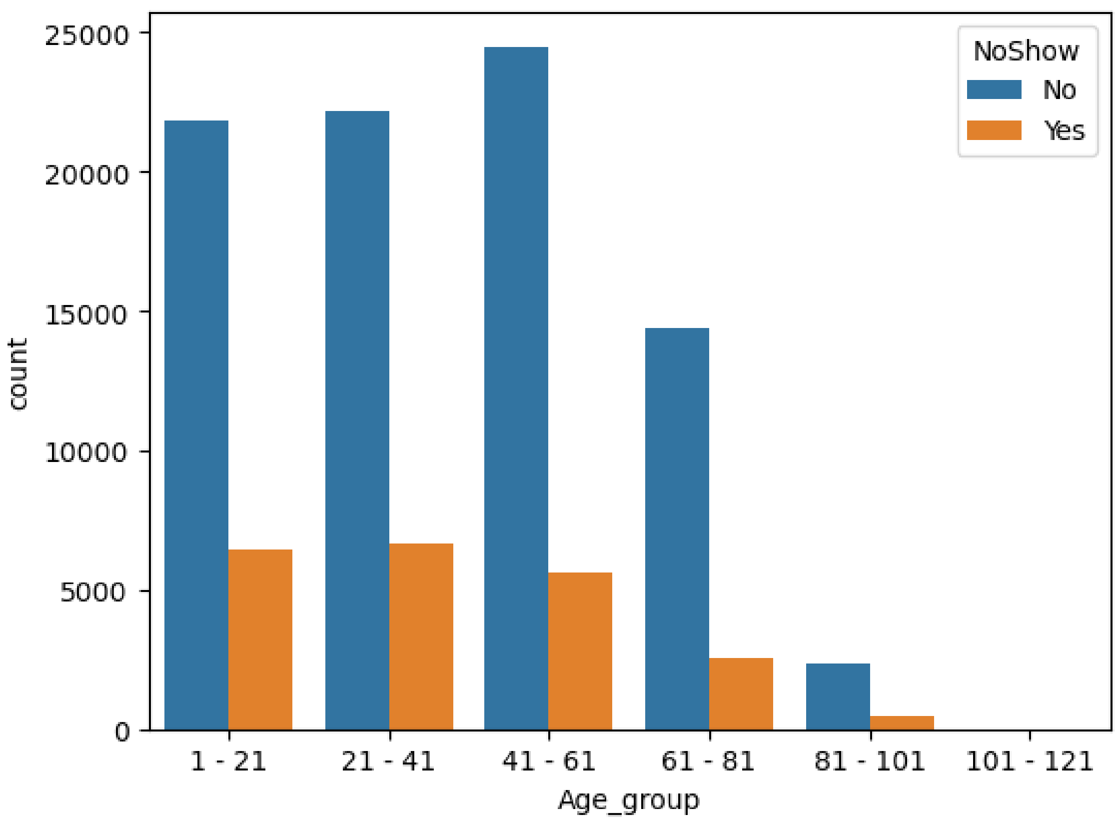

NoShow
No
Yes
count
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Alcoholism

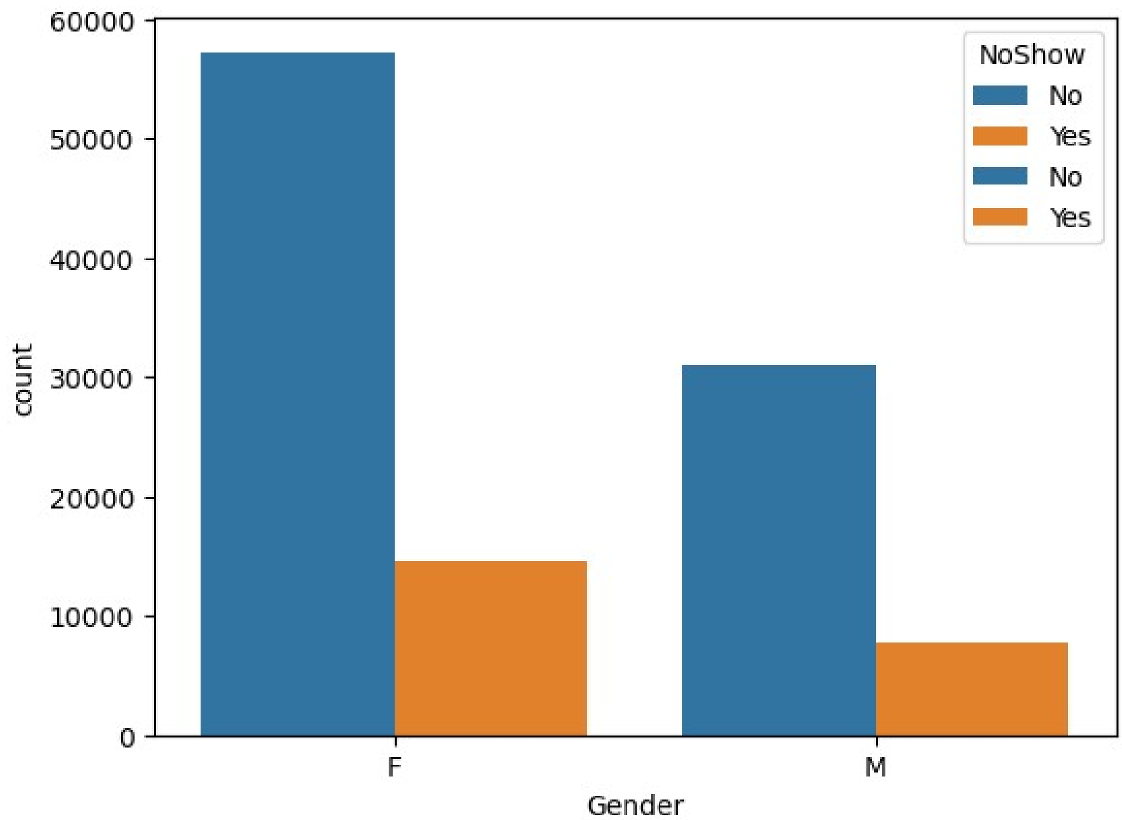

NoShow
No
Yes
count
80000
60000
40000
20000
0
0
1
2
3
4
Handicap


NoShow
No
Yes
count
60000
50000
40000
30000
20000
10000
0
0
1
SMSReceived

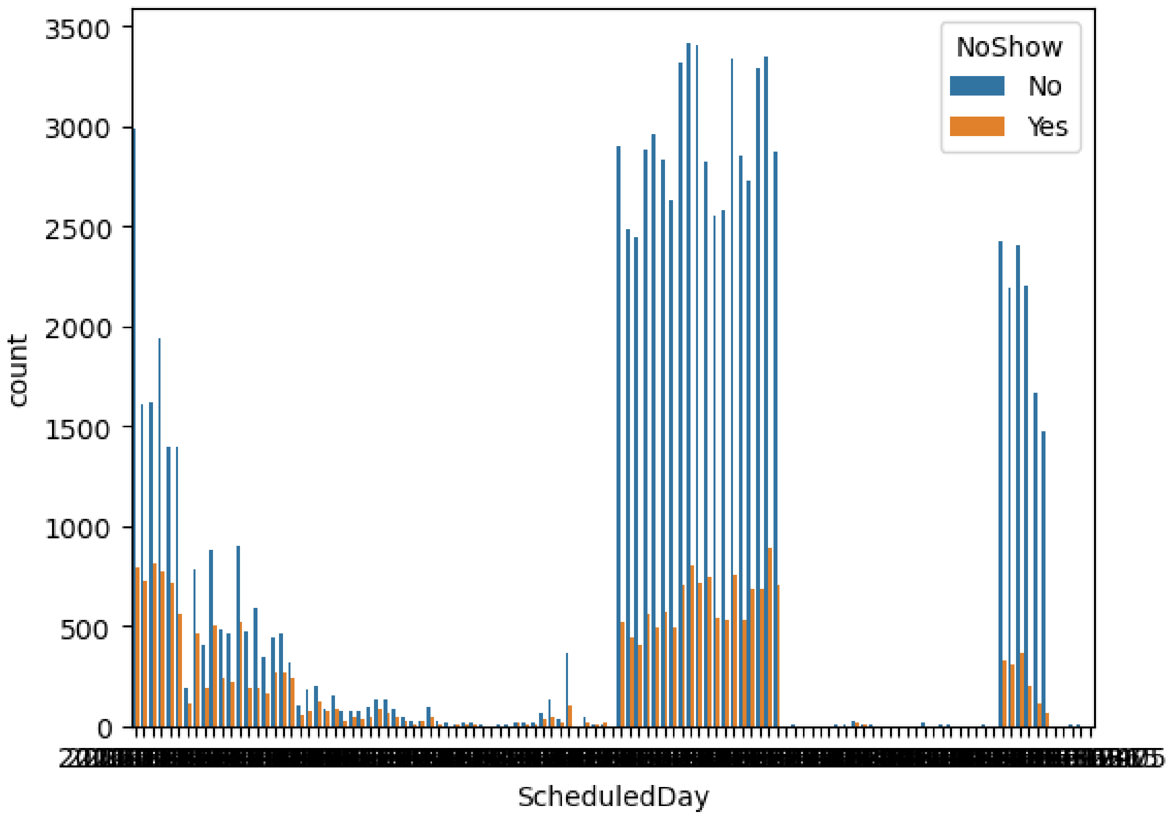

NoShow
Yes
No
count
20000
17500
15000
12500
10000
7500
5000
2500
0
0
1
2
3
4
5
s_wday


NoShow
No
Yes
count
20000
17500
15000
12500
10000
7500
5000
2500
0
0
1
2
3
4
5
ap_wday


NoShow
No
Yes
count
25000
20000
15000
10000
5000
0
1 - 21
21 - 41
41 - 61
61 - 81
81 - 101
101 - 121
Age_group

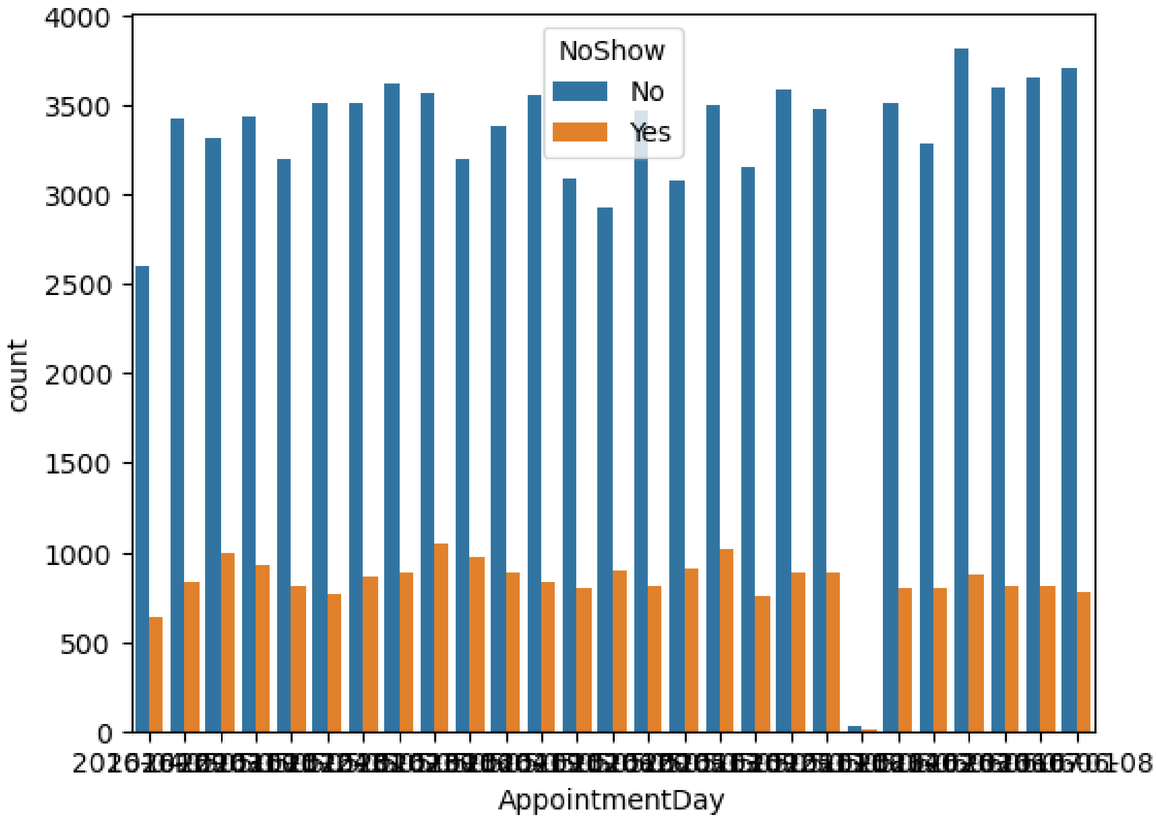



```
---------- ScheduledDay ----------
ScheduledDay
2016-05-03    4238
2016-05-02    4216
2016-05-16    4120
2016-05-05    4095
2016-05-10    4024
              ...
2016-04-16       1
2016-01-28       1
2015-11-10       1
2016-03-19       1
2016-03-05       1
Name: count, Length: 111, dtype: int64
```

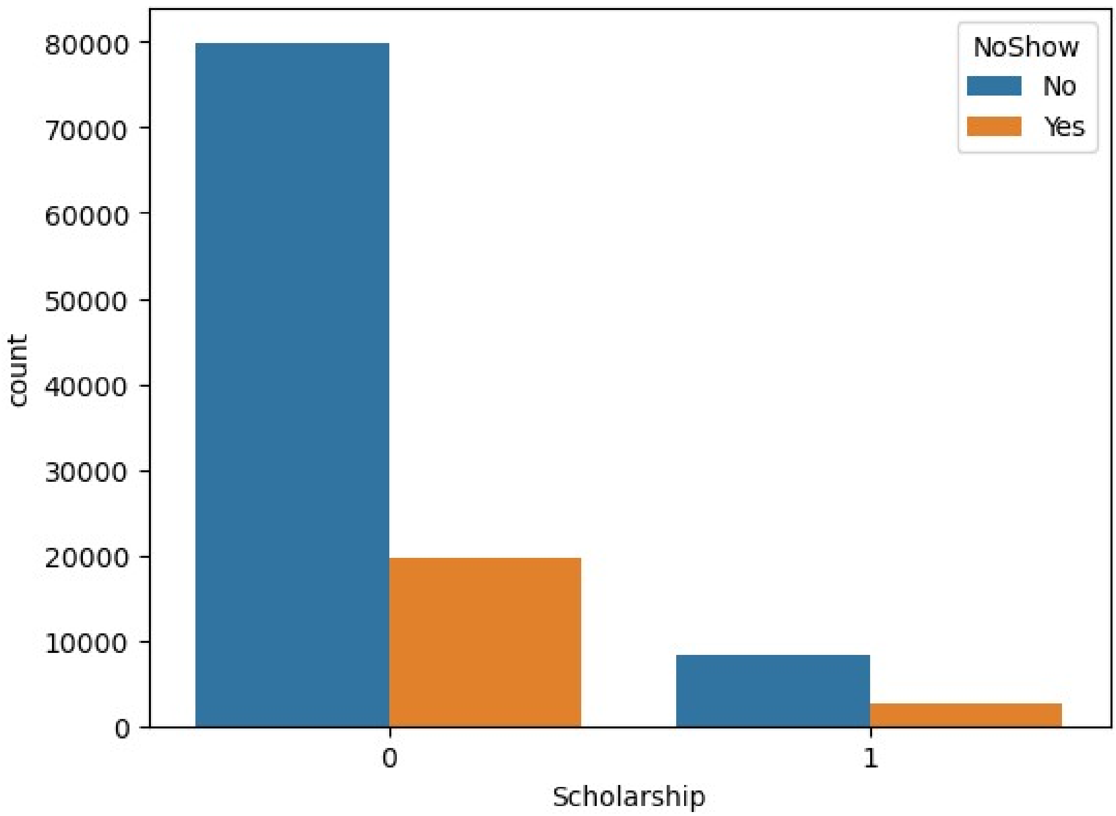



```
---------- AppointmentDay ----------
AppointmentDay
2016-06-06    4692
2016-05-16    4613
2016-05-09    4520
2016-05-30    4514
2016-06-08    4479
2016-05-11    4474
2016-06-01    4464
2016-06-07    4416
2016-05-12    4394
2016-05-02    4376
2016-05-18    4373
2016-05-17    4372
2016-06-02    4310
2016-05-10    4308
2016-05-31    4279
2016-05-05    4273
2016-05-19    4270
2016-05-03    4256
2016-05-04    4168
2016-06-03    4090
2016-05-24    4009
2016-05-13    3987
```

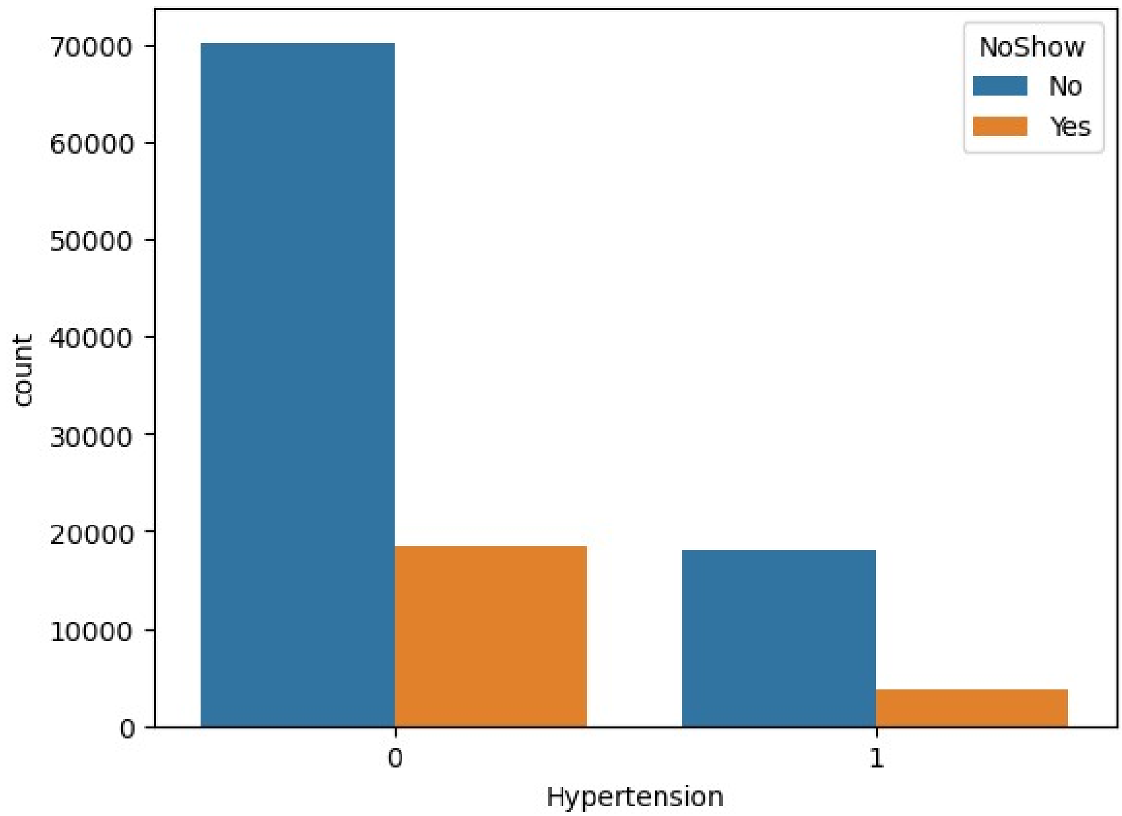
```
2016-05-25    3909
2016-05-06    3879
2016-05-20    3828
2016-04-29    3235
2016-05-14      39
Name: count, dtype: int64
```



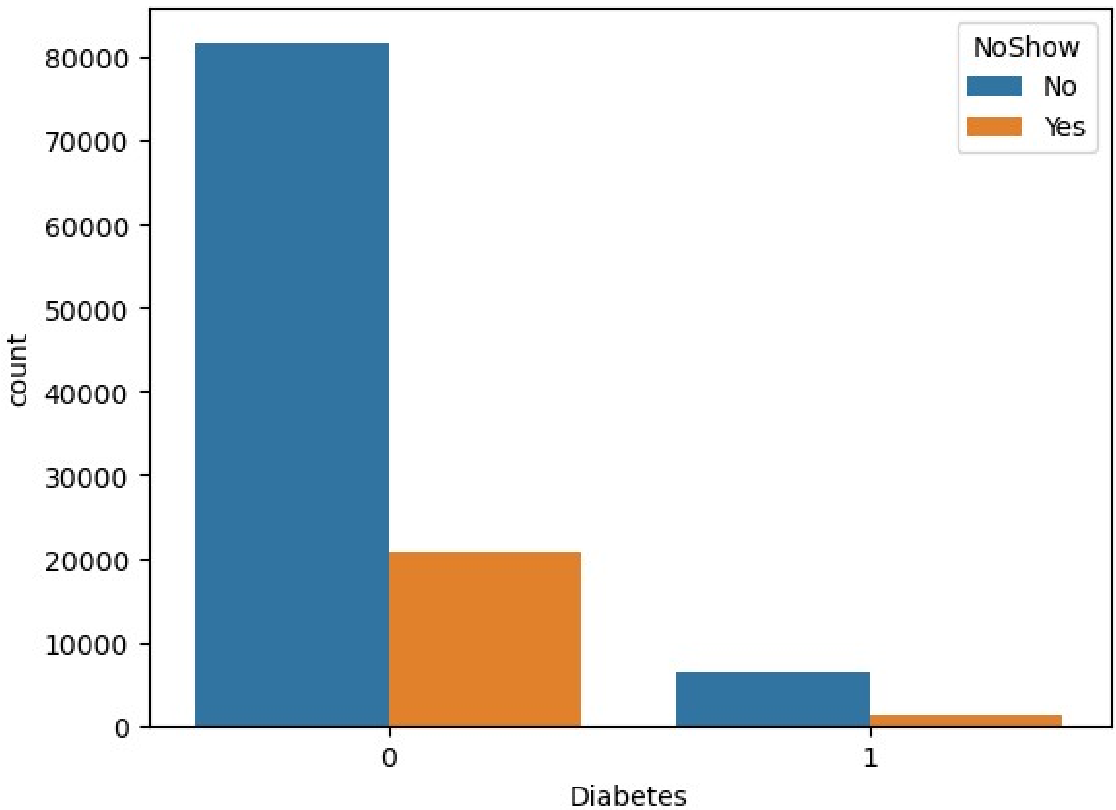

```
---------- Scholarship ----------
Scholarship
0    99666
1    10861
Name: count, dtype: int64
```



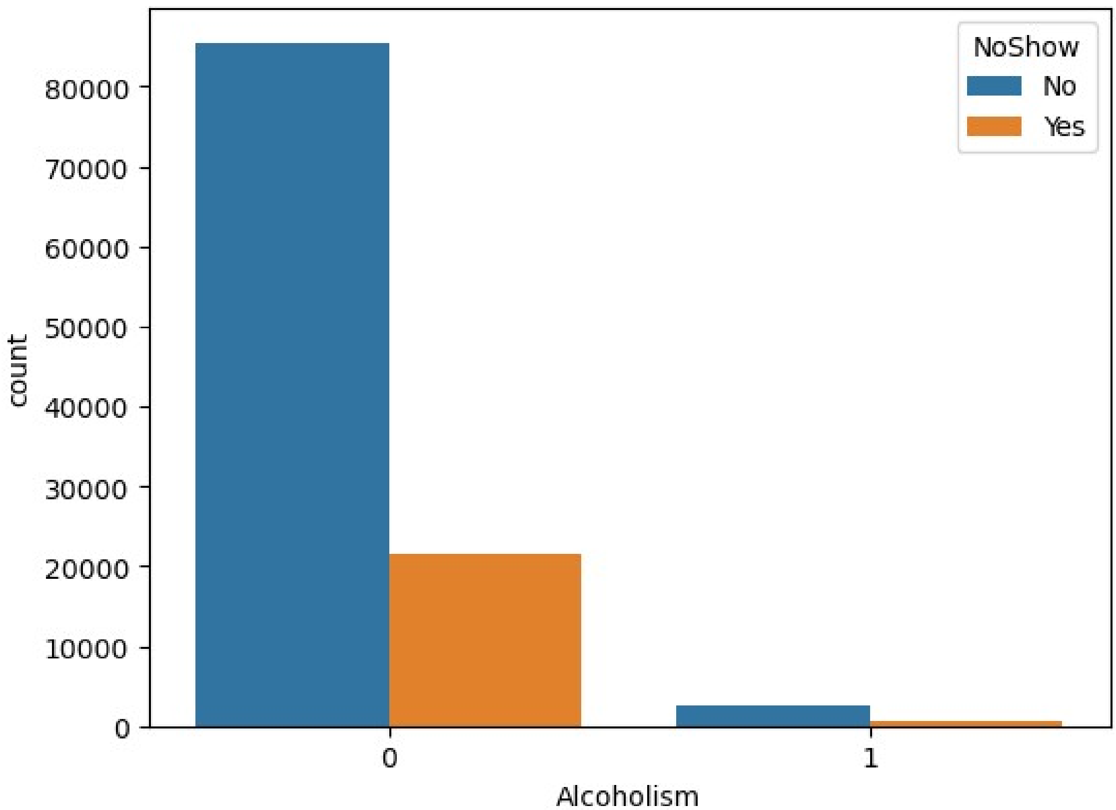

```
---------- Hypertension ----------
Hypertension
0    88726
1    21801
Name: count, dtype: int64
```



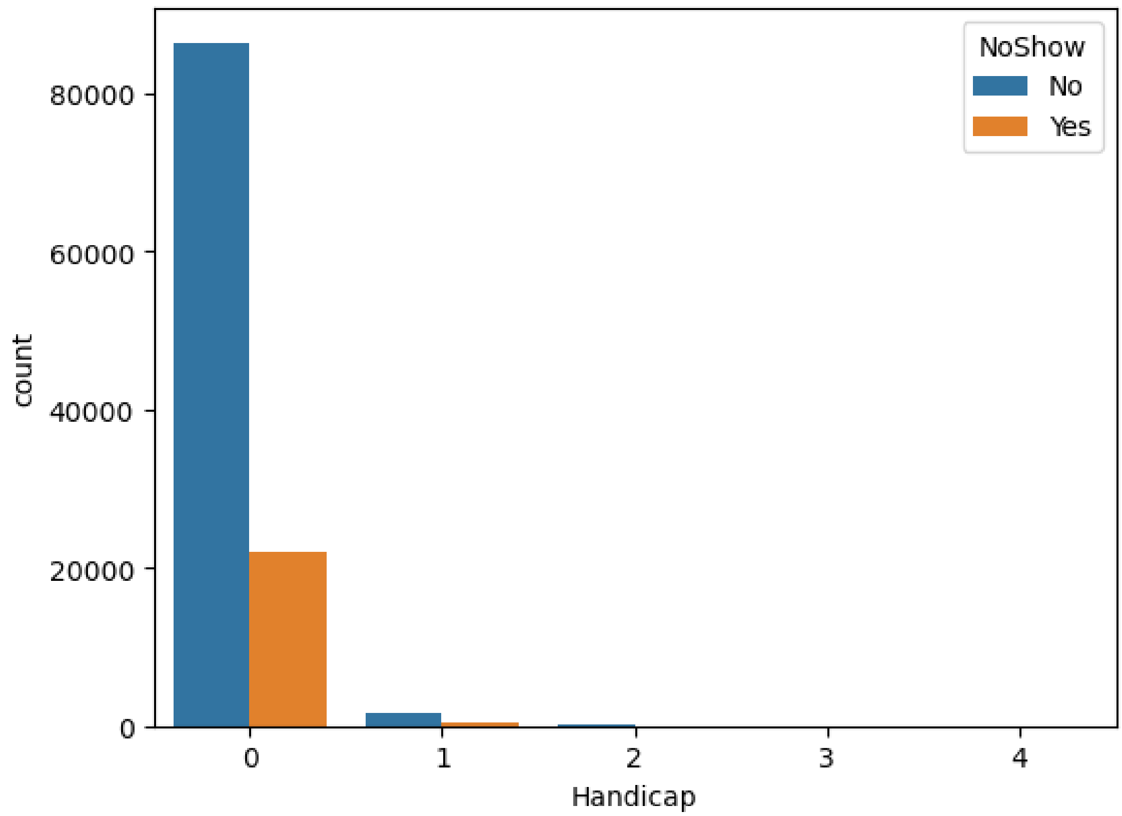

```
---------- Diabetes ----------
Diabetes
0    102584
1      7943
Name: count, dtype: int64
```



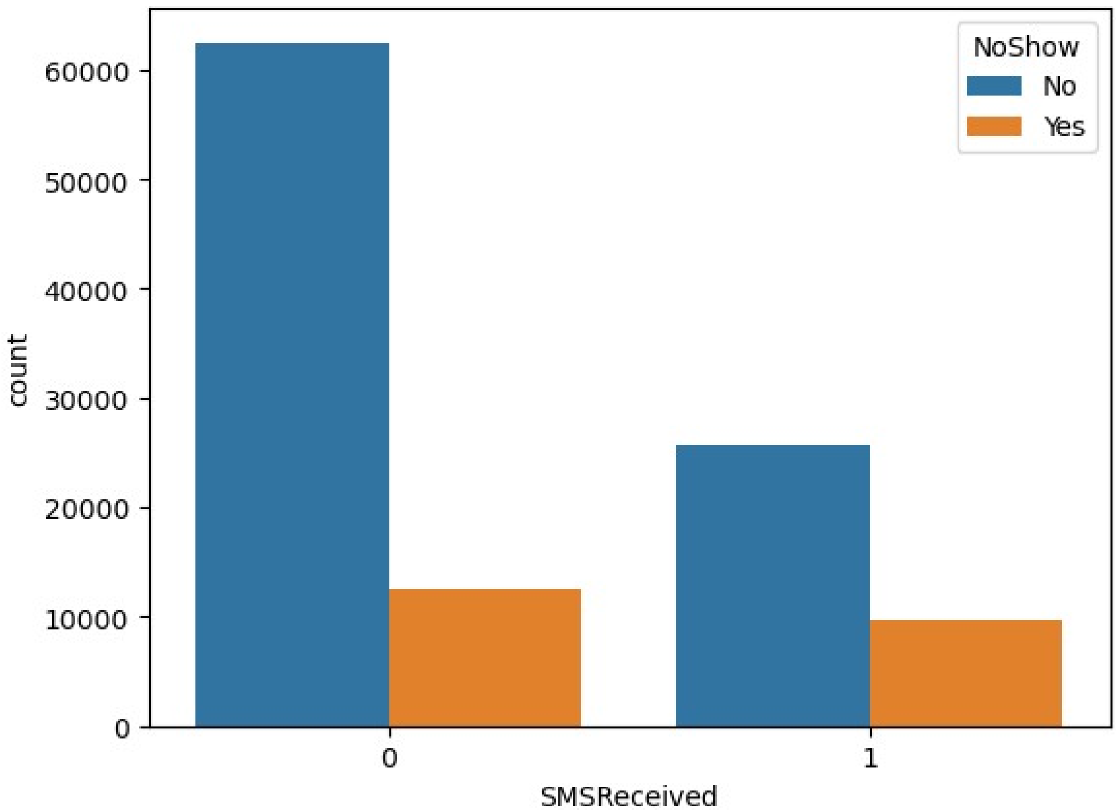

```
---------- Alcoholism ----------
Alcoholism
0    107167
1      3360
Name: count, dtype: int64
```



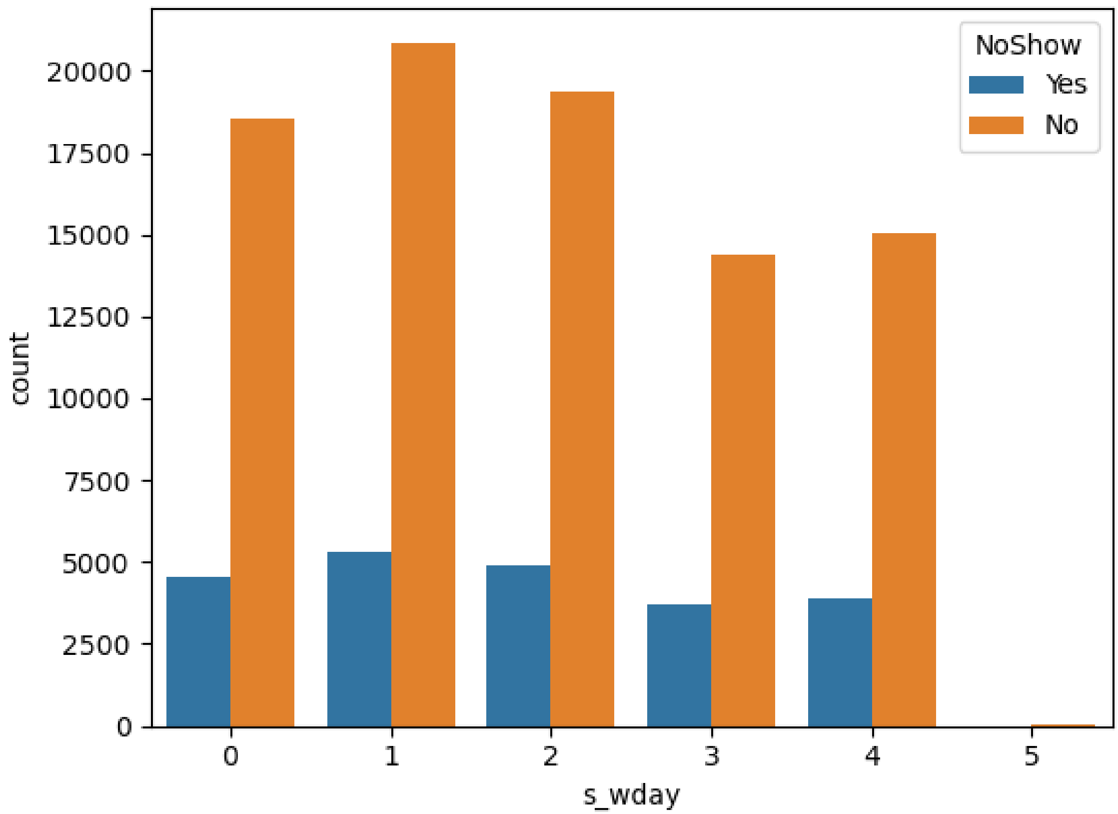

```
---------- Handicap ----------
Handicap
0    108286
1      2042
2       183
3        13
4         3
Name: count, dtype: int64
```



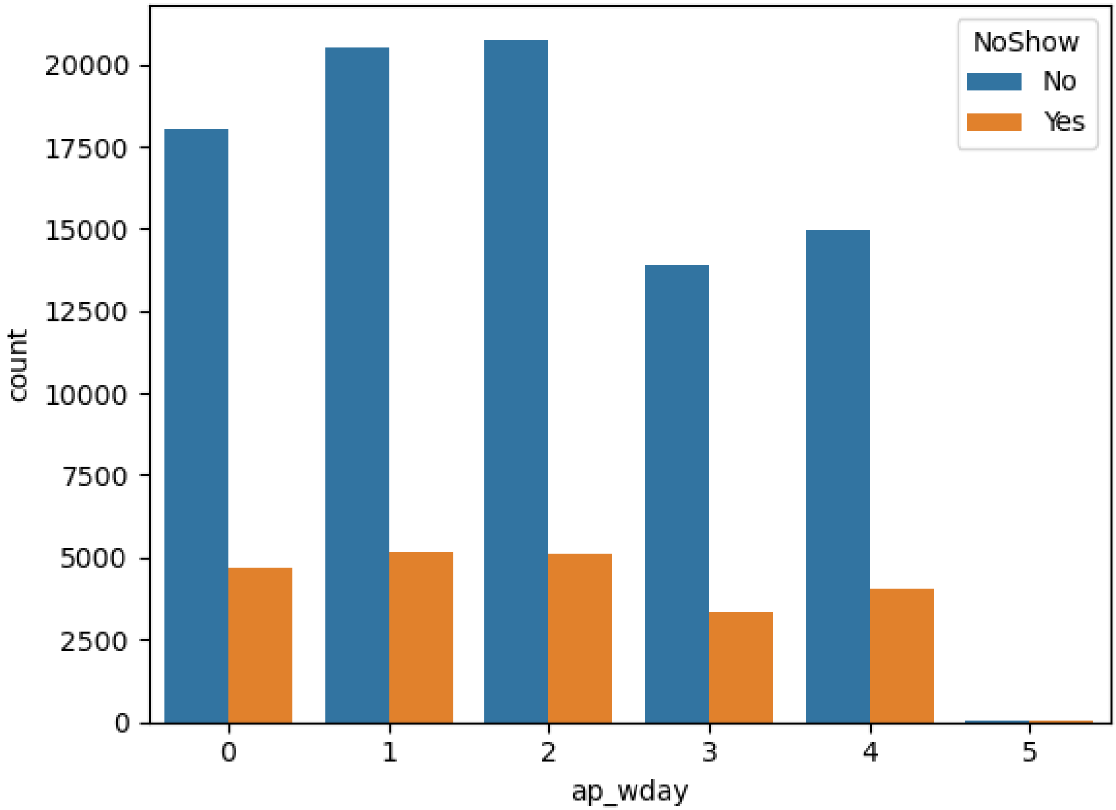

```
---------- SMSReceived ----------
SMSReceived
0    75045
1    35482
Name: count, dtype: int64
```



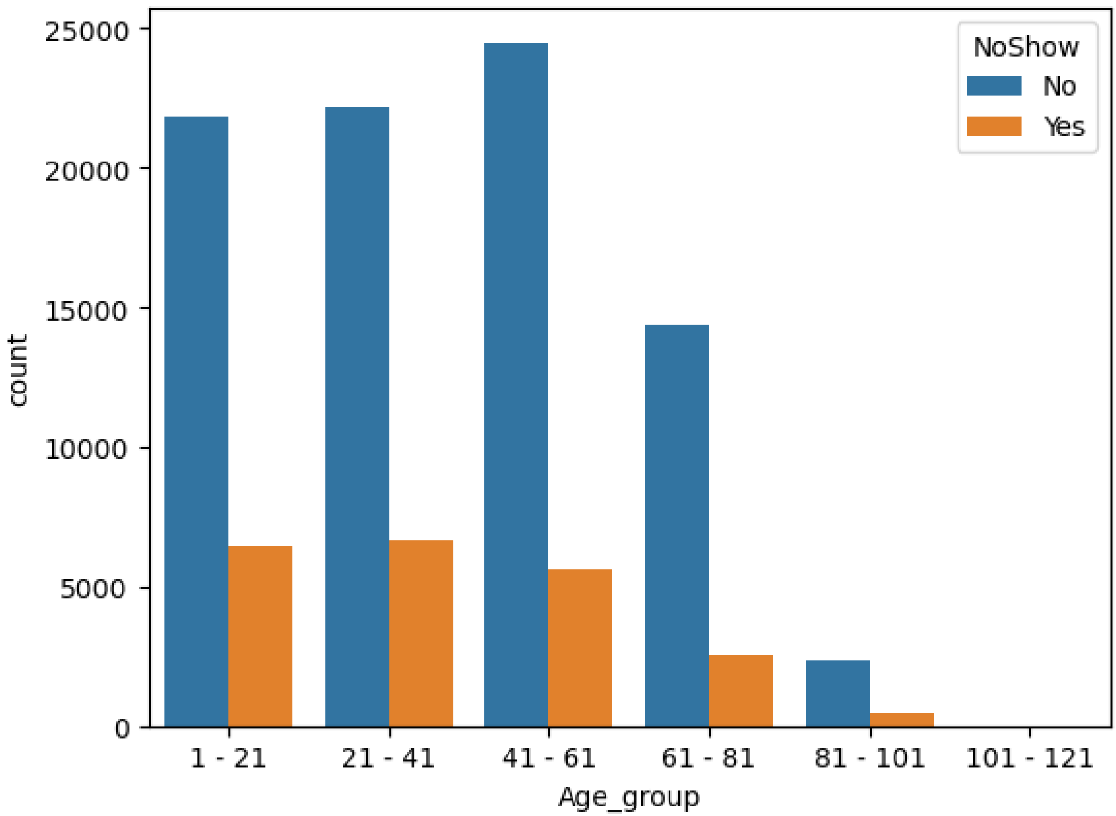

```
---------- s_wday ----------
s_wday
1    26168
2    24262
0    23085
4    18915
3    18073
5       24
Name: count, dtype: int64
```




```
---------- ap_wday ----------
ap_wday
2    25867
1    25640
0    22715
4    19019
3    17247
5       39
Name: count, dtype: int64
```




```
---------- Age_group ----------
Age_group
41 - 61      30081
21 - 41      28835
1 - 21       28309
61 - 81      16910
81 - 101      2845
101 - 121        7
Name: count, dtype: int64
```




```
data['NoShow'] = np.where(data.NoShow == 'Yes',1,0)

data.NoShow.value_counts()

NoShow
0    88208
1    22319
Name: count, dtype: int64
```

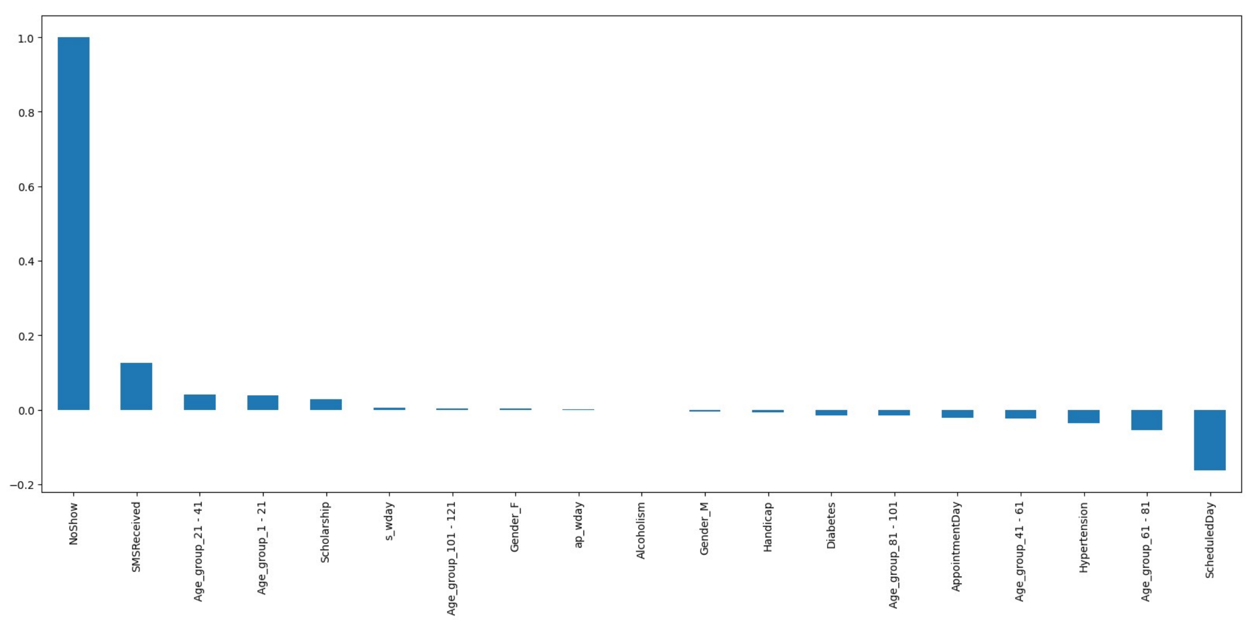
## Converting categorical values into dummy variables

```
data_dummies = pd.get_dummies(data)
data_dummies.head()

  ScheduledDay AppointmentDay  Scholarship  Hypertension  Diabetes  \
0   2016-04-29     2016-04-29            0             1         0
1   2016-04-29     2016-04-29            0             0         0
2   2016-04-29     2016-04-29            0             0         0
3   2016-04-29     2016-04-29            0             0         0
4   2016-04-29     2016-04-29            0             1         1

   Alcoholism  Handicap  SMSReceived  NoShow  s_wday  ap_wday
Gender_F  \
0           0         0            0       0       4        4
True
```

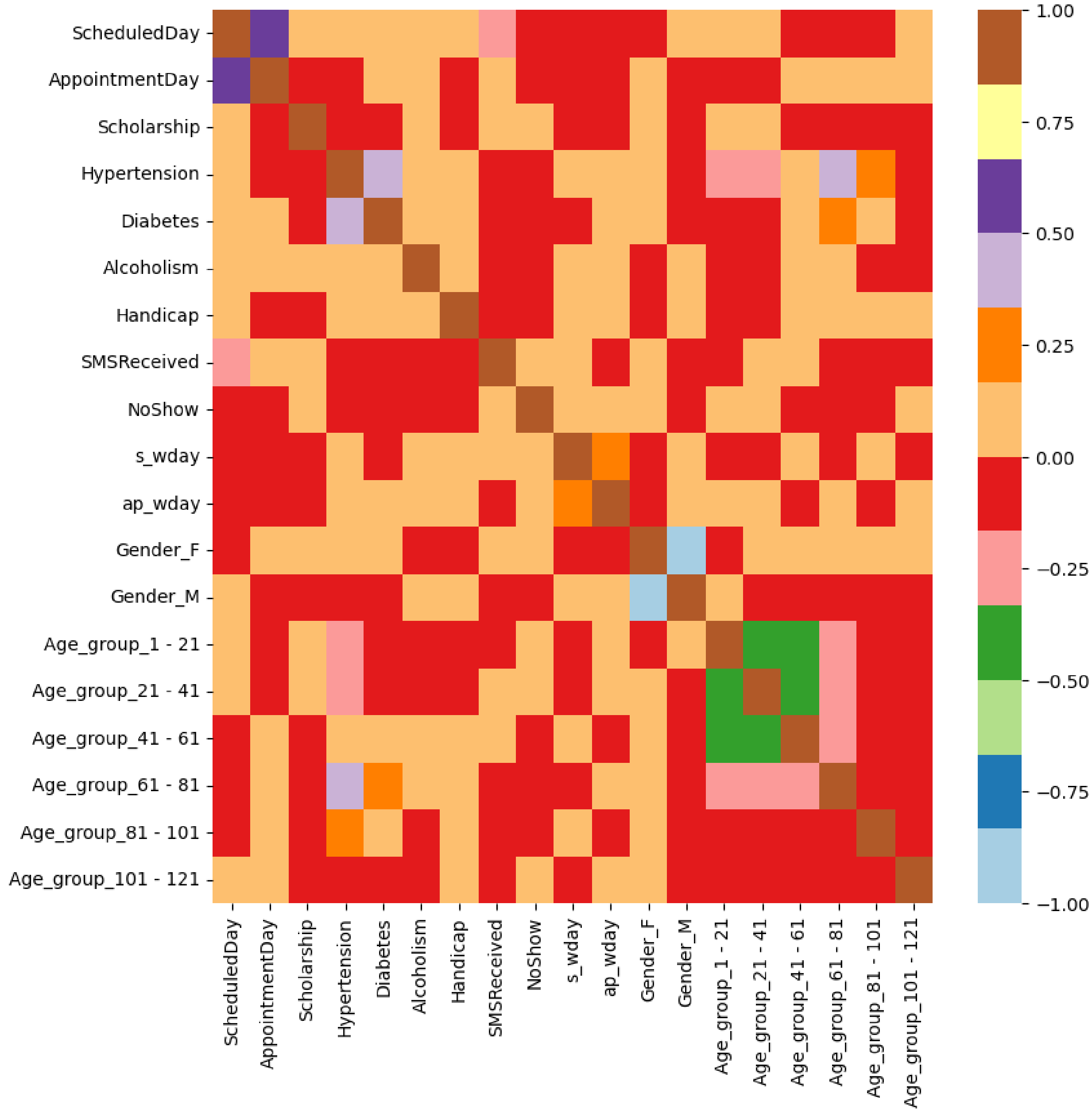
```
1            0         0            0       0       4        4
False
2            0         0            0       0       4        4
True
3            0         0            0       0       4        4
True
4            0         0            0       0       4        4
True

   Gender_M  Age_group_1 - 21  Age_group_21 - 41  Age_group_41 - 61  \
0     False             False              False              False
1      True             False              False               True
2     False             False              False              False
3     False              True              False              False
4     False             False              False               True

   Age_group_61 - 81  Age_group_81 - 101  Age_group_101 - 121
0               True               False                False
1              False               False                False
2               True               False                False
3              False               False                False
4              False               False                False
```

# Build a Correlation of all predictors with No-show

```
plt.figure(figsize=(20,8))
data_dummies.corr()['NoShow'].sort_values(ascending =
False).plot(kind='bar')
plt.show()
```



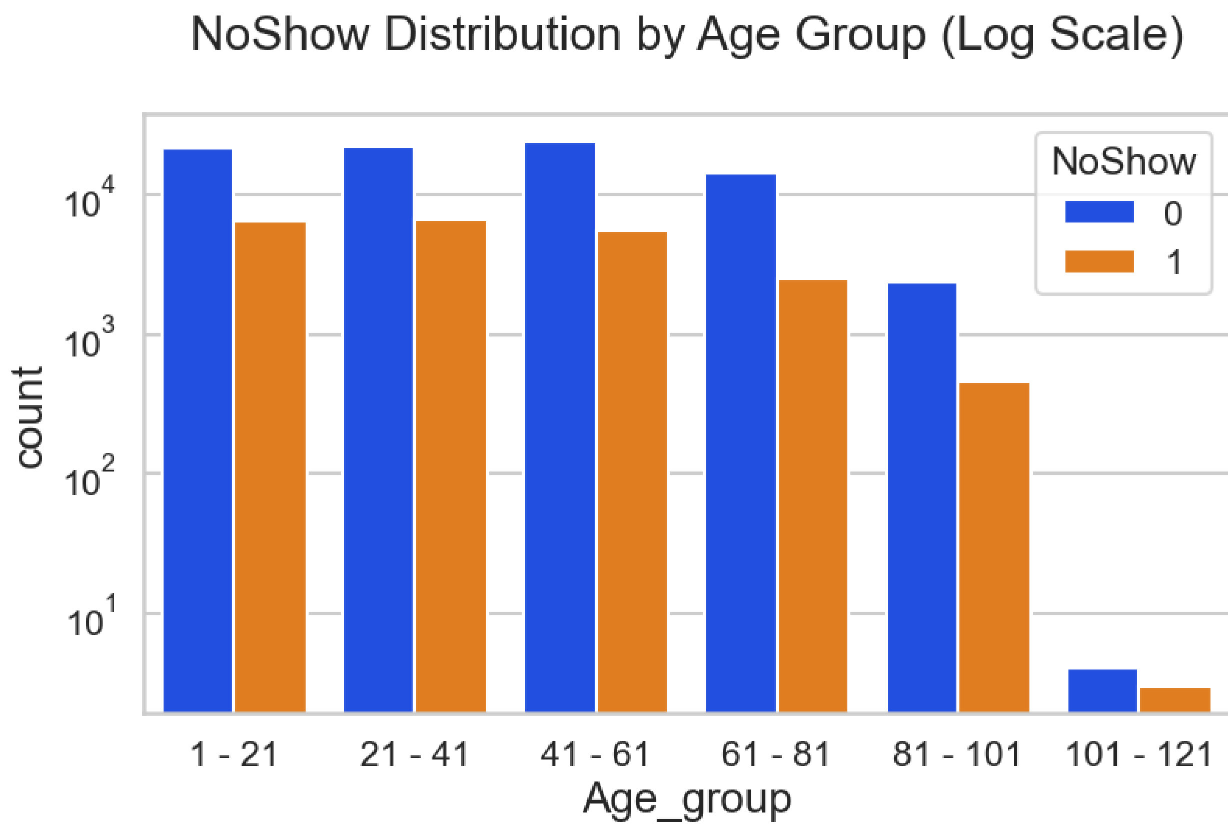
```
plt.figure(figsize=(9,9))
sns.heatmap(data_dummies.corr(), cmap="Paired")
plt.show()
```




## Bivariate Analysis

```
new_df1_target0=data.loc[data["NoShow"]==0]
new_df1_target1=data.loc[data["NoShow"]==1]

plt.figure(figsize=(9, 5))
ax = sns.countplot(data=data, x='Age_group', hue='NoShow',
```

```
palette='bright')
ax.set_yscale("log")
plt.title("NoShow Distribution by Age Group (Log Scale)")
plt.show()
```



# FINDINGS:-

1. Female patients have taken more appointments then male patients

2. Ratio of Nohow and Show is almost equal for age group except Age 0 and Age 1 with 80% show rate for each age group

3. Each Neighbourhood have almost 80% show rate

4. There are 99666 patients without Scholarship and out of them around 80% have come for the visit and out of the 21801 patients with Scholarship around 75% of them have come for the visit.

5. Tthere are around 88,726 patients without Hypertension and out of them around 78% have come for the visit and Out of the 21801 patients with Hypertension around 85% of them have come for the visit.

6. There are around 102,584 patients without Diabetes and out of them around 80% have come for the visit and Out of the 7,943 patients with Diabetes around 83% of them have come for the visit.

7. There are around 75,045 patients who have not received SMS and out of them around 84% have come for the visit and out of the 35,482 patients who have received SMS around 72% of them have come for the visit.

8. There is no appointments on sunday and on saturday appointments are very less in comparision to other week days